UM-3 field equations

The complete set of coupled field equations of Einstein-Lovelock gravity, dirac16complex, the Higgs doublet with the ad hoc vacuum, the gauge fields of SU(3), SU(2) and U(1), and three generations of quarks and leptons with three right-handed neutrinos

Underlying theory: Patrick L. Nash Derived and verified by Claude (Anthropic)

Field equations of UM-3, 2026-10-24

Repository once-ere/Dirac_claude, folder universe_model_3

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1 How to read this document

1.1 What this document contains

This document states the complete set of coupled field equations of the universe model UM-3, derives every one of them from one action, and explains every term. It is self-contained: every equation it uses is written in it, and every symbol of every equation is defined right after the equation. The sections are:

- **Where the fields live** (this section): the author's metric has a fold, the seven-dimensional surface $z = \pi/2$, that carries energy; the Standard Model lives on the fold, dirac16complex and gravity in the bulk.
- **The action** (IN-1 to IN-16): every term of the action, written out and explained.
- **Gravity** (FE-1 to FE-11): the Einstein-Lovelock equations in the bulk, the junction conditions at the fold, the energy-momentum tensor of every field and the conservation laws.
- **The matter fields** (FE-12 to FE-26): the field equations of dirac16complex, of the Higgs doublet, of the gauge fields and of the quarks and leptons, and the equations of the ad hoc Higgs vacuum.
- **The background** (FE-27 to FE-35): the author's metric, its exact curvature, the exact background equations in the bulk and at the fold, the author's vacuum, the static late-time state, the frozen extra times and the power series of the metric.
- **dirac16complex: modes and Kohn-Sham equations** (FE-36 to FE-43): the equation of the hidden direction, the bulk continuum, the states bound to the fold, the Kohn-Sham equations with the mean fields of every other field, the Krein norm.
- **The Standard Model on the fold** (FE-44 to FE-47): the four-dimensional couplings, the masses with the ad hoc vacuum, the mixing matrices and the four-dimensional couplings of the bound states of dirac16complex.
- **The three neutrino flavours** (FE-48 to FE-53): the flavour evolution in vacuum, in matter, in dense neutrino gases (supernovae) and in the early Universe.
- **The cosmological perturbations** (FE-54 to FE-58): the metric as a power series and the Einstein-Boltzmann equations of the CMB, the CNB and the BAO.
- **The thermal history** (FE-59 to FE-62): the expansion of the fold, the degrees of freedom and the Boltzmann equations of the bound states of dirac16complex and of the right-handed neutrinos.
- **The coupled system**: which field appears in which equation, and how the system is solved.
- **What was verified**: the checks of the computations behind every derived statement.

1.2 The status labels

Every equation and every result carries one of the labels

- **AUTHOR**: from the author (his notebooks `4+4.nb` and `Dirac_Einstein-Lovelock_4+4_Nash.nb`, and his order of 2026-10-24: the coordinates, the metric, the Lovelock weights, the Dirac matrices, the chiral projectors, the list of fields, the Higgs potential and the ad hoc vacuum);
- **RECORD**: from the record of this repository (the Lagrangian of dirac16complex with its canonical spin connection and its contact self-interaction, the Kohn-Sham treatment);
- **STANDARD**: textbook physics;
- **ASSUMED**: a choice made in UM-3 without derivation;
- **INVENTED**: a new interaction of UM-3 (the portals of dirac16complex);
- **DERIVED**: computed and verified in UM-3 (the verification is stated);
- **ESTIMATE**: an order-of-magnitude evaluation;
- **OPEN**: not settled.

1.3 Every equation is followed by its symbols

Every display equation is followed by the list "Symbols in (FE-n)" (or "Symbols in this equation" for a display without a tag) of EVERY symbol that appears in it: every quantity, field, coupling, constant and operator, every index letter with the values it takes, every mark (prime, dagger, transpose, star), the imaginary unit, Euler's number and the differential. Each item gives the meaning, the unit (natural units $\hbar = c = 1$), the value where one exists, and the status. A field equation carries the tag FE-1 to FE-62 and a term of the action the tag IN-1 to IN-16; the

numbering is that of the inventory of UM-3 and is the same in every UM-3 document. Displays without a tag are steps of a derivation.

1.4 Coordinates, signature and units

The author's coordinates are $X = \{x_0, x_1, \ldots, x_7\}$, always with the hidden direction first: x_0 is the hidden space-like coordinate, on which the metric depends periodically with the period $\pi/(6H)$ through the angle $z = 6Hx_0$; x_1, x_2, x_3 are ordinary space, x_4 is the time, and x_5, x_6, x_7 are three extra time-like coordinates. The signature is $(+,+,+,+,-,-,-,-)$ in this order; the four observable coordinates $x_1, \ldots, x_4$ have the signature $(+,+,+,-)$. H is the author's constant (an inverse length); this document keeps it as a free parameter. Units are natural, $\hbar = c = 1$; in eight dimensions the field dirac16complex has the dimension $\mathrm{GeV}^{7/2}$ and the gravitational coupling $\kappa = 8\pi G_8$ the dimension GeV^{-6}. A prime on a function of the time x_4 alone is its derivative $\mathrm{d}/\mathrm{d}x_4$ (in the cosmological perturbation equations, and only there, a dot is the derivative with respect to the conformal time). The proper coordinate of the hidden direction is σ (defined in FE-27): $\sigma = 0$ is the fold $z = \pi/2$.

1.5 The fields of UM-3 and where they live

The author orders a model universe with the following coupled fields (his order of 2026-10-24, items 1 to 10); UM-3 adds the right-handed quark singlets, the Yukawa couplings and three right-handed neutrinos (accepted by the author on 2026-10-24 at 17:32), the stabiliser of the bulk, the fold fluid and the induced gravity of the fold (ASSUMED, explained below).

field	symbol	what it is	lives in	it couples to	field equations
Einstein-Lovelock gravity	g_{MN}	the metric of eight dimensions	everywhere	every field	FE-1 to FE-11, FE-27 to FE-35, FE-54 to FE-56
dirac16complex	Ψ	the author's 16-component complex spinor field; its left- and right-handed halves Ψ_L, Ψ_R	the bulk	gravity (spin connection), itself, every Standard-Model field at the fold (IN-16)	FE-6, FE-12 to FE-14, FE-36 to FE-43, FE-47
Higgs doublet	Φ	the scalar doublet with $V = \mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2$, expanded about the ad hoc vacuum $\Phi = (\mathcal{A}, v + h)/\sqrt{2}$	the fold	gravity, W, B, the fermions (Yukawa), Ψ_L only	FE-7, FE-15, FE-26, FE-45
three left-handed lepton doublets	ℓ_i	$(\nu_{\mathrm{L}i}, e_{\mathrm{L}i})$, $i = 1, 2, 3$	the fold	gravity, W, B, Φ, Ψ_L and Ψ_R	FE-9, FE-22, FE-48 to FE-53
right-handed electron, muon, tau	$e_{\mathrm{R}i}$	the charged-lepton singlets	the fold	gravity, B, Φ, Ψ_L, Ψ_R	FE-9, FE-23
three right-handed neutrinos	$\nu_{\mathrm{R}i}$	gauge singlets; Dirac neutrino masses	the fold	gravity, Φ, Ψ_L, Ψ_R	FE-9, FE-24, FE-62
gauge fields of $SU(2)_L$ and $U(1)_Y$	W^1, W^2, W^3, B	the electroweak gauge fields	the fold	gravity, Φ, the fermions, Ψ_L, Ψ_R	FE-8, FE-17, FE-18
three quark generations	Q_i, $u_{\mathrm{R}i}$, $d_{\mathrm{R}i}$	the quark doublets and singlets	the fold	gravity, gluons, W, B, Φ, Ψ_L, Ψ_R	FE-9, FE-19 to FE-21
gluons	G^A	the gauge field of $SU(3)_C$	the fold	gravity, quarks, Ψ_L, Ψ_R	FE-8, FE-16

field	symbol	what it is	lives in	it couples to	field equations
stabiliser	$\mathcal{T}_{\text{stab}}$	a medium of the bulk that keeps the late-time bulk static	the bulk	gravity	FE-3, FE-32
fold fluid	$\mathcal{S}_{\text{ff}}$	the thin source that the kink of the metric requires at the fold	the fold	gravity	FE-4, FE-5, FE-31, FE-32
induced gravity of the fold	κ_4	an Einstein-Hilbert term of the fold	the fold	gravity	FE-5, FE-59

Why the Standard Model lives on the fold (DERIVED). In the proper coordinate σ of the hidden direction (FE-27), one period of x_0 is the whole line $-\infty < \sigma < \infty$ and the author's metric is $\mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\big(\mathrm{e}^{2a_4}\,\mathrm{d}\vec{x}^2 - \mathrm{e}^{-2a_4}b_4^2\,\mathrm{d}\vec{x}_5^2\big) - \mathrm{d}x_4^2$: ordinary space is warped by $\mathrm{e}^{-H|\sigma|}$, the time x_4 is not. A field confined to the fold $\sigma = 0$ has the light cone of the fold; a field that spreads into the bulk moves along ordinary space with a speed set by the σ-average of $\mathrm{e}^{2H|\sigma|}$ over its profile. With the author's metric a bulk photon moves $\sqrt{2}$ times faster than light on the fold, a bulk scalar $\sqrt{3/2}$ times, a bulk fermion with a profile of width $1/M$ $2M/(2M-H)$ times (DERIVED, `um3_bulk.py`, checks B1 to B3). Species with different profiles would have different limiting speeds (excluded at the level 10^{-15} to 10^{-20} for photons, electrons and nucleons), and three neutrino flavours with three profiles would oscillate with energy-proportional phases that observation excludes. The author's request for correct flavour conversion therefore fixes the place of the Standard Model: all of it lives on the fold, where it has one light cone, and its flavour physics is that of the Standard Model with three Dirac neutrinos. dirac16complex is the author's field of eight dimensions and lives in the bulk; it couples to the Standard Model at the fold.

Why the fold carries a source (DERIVED). The warp $\mathrm{e}^{-2H|\sigma|}$ has a kink at the fold. The Einstein-Lovelock equations, which are linear in the second derivatives of the metric along σ, then contain a delta function at $\sigma = 0$: the metric is a solution only if the fold carries a thin source $\mathcal{S}^a{}_b$ (FE-4). At the author's vacuum $\kappa\mathcal{S} = \mathrm{diag}(-10H, -6H, -10H)$ for (ordinary space, time, extra times) (DERIVED, `um3_fold.py`, check J5). This source is the fold fluid together with the Standard Model and the portals.

Why the fold has its own Einstein-Hilbert term (ASSUMED). The transverse traceless gravitons of the bulk, with the author's Lovelock weights, propagate with $c_{\text{g}}^2 = 15/2$ times the light cone of the fold at the static late-time state, and they have no time-kinetic term at all at the author's vacuum (DERIVED, `um3_bulk.py`, B5). Gravitational waves move with the speed of light to 10^{-15} (GW170817). UM-3 therefore gives the fold an induced Einstein-Hilbert term (IN-9) that dominates four-dimensional gravity; the bulk contributes a fraction $r \lesssim 10^{-15}$ (FE-59).

1.6 Conventions

- Indices. M, N, P, Q (with numbered copies $M_1, M_2, \ldots$): coordinate indices of eight dimensions; a, b, c, d: frame indices of eight dimensions (the frame direction (0) is the σ direction); μ, ν, ρ: coordinate indices of the four observable directions $x_1, \ldots, x_4$; $A, B, C = 1, \ldots, 8$: colour (adjoint) indices; $I, J, K = 1, 2, 3$: weak-isospin (adjoint) indices; $i, j = 1, 2, 3$: generations (and the neutrino mass states); $\alpha, \beta = e, \mu, \tau$: lepton flavours; $p = 1, 2, 3$: the Lovelock order; $X = \mathrm{L}, \mathrm{R}$: the two halves of dirac16complex; $f \in \{Q, u, d, \ell, e, \nu\}$: the six kinds of Standard-Model multiplets. A repeated index is summed over its range.
- Two kinds of Dirac matrices. The author's real 16×16 matrices T_a (IN-5) act on dirac16complex; the four-dimensional complex 4×4 matrices $\hat{\gamma}^\mu$ (IN-14) act on the Standard-Model fermions. They are never multiplied with each other; the couplings between dirac16complex and the Standard Model are products of Lorentz tensors.
- Normalisation of the fold. No field depends on the extra times x_5, x_6, x_7 (ASSUMED: a dependence on them gives tachyonic four-dimensional modes, FE-38). Integrated over them, the extra times give their physical volume on the fold, V_*; the Standard-Model fields are normalised as four-dimensional fields, the couplings are their measured four-dimensional values, and the fold action carries the factor $1/V_*$ (IN-8).
- A word or a particle name in a subscript is upright (m_{W}, ν_{R}, Ψ_{L}); an italic letter in a subscript is an index. e is Euler's number, i the imaginary unit, d the differential; δ is the Kronecker delta and δ_{D} the Dirac delta function.

2 The action

All field equations of UM-3 follow from one action by varying it with respect to every field. This section writes out every term of the action (IN-1 to IN-16) and says in plain words what it couples. The Standard-Model part has exactly the gauge group, the fermion content (three generations, each with a right-handed neutrino), the Higgs potential and the gauge couplings of the Standard Model; it lives on the fold. dirac16complex is a singlet of the Standard-Model gauge group; its left-handed and right-handed halves couple to every Standard-Model field at the fold through gauge-invariant portals with their own coupling constants, and only its left-handed half couples to the Higgs doublet.

2.1 The sum of all terms

$$\begin{aligned} S_{\mathrm{tot}} &= S_{\mathrm{EL}} + S_{\mathrm{stab}} + S_{\Psi} + S_{\mathrm{fold}}, \\ S_{\mathrm{fold}} &= \int \mathrm{d}^8X\, \sqrt{|g|}\, \delta_{\mathrm{D}}(\sigma)\, \frac{1}{V_*} \left(\mathcal{L}_{\mathrm{IG}} + \mathcal{L}_{\mathrm{ff}} + \mathcal{L}_{\mathrm{H}} + \mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_{\mathrm{f}} + \mathcal{L}_{\mathrm{Y}} + \mathcal{L}_{\mathrm{P}} \right) \end{aligned} \tag{IN-1}$$

Symbols in (IN-1).

- S_{tot}: The action of UM-3 (the integral of the Lagrangian density over spacetime); its variation with respect to every field gives the field equations. Unit: dimensionless ($\hbar = 1$). Status: ASSUMED.
- S_{EL}: The Einstein-Lovelock action of gravity in eight dimensions. Unit: dimensionless. Status: AUTHOR.
- S_{stab}: The action of the stabiliser of the bulk (a medium specified by its energy-momentum tensor $\mathcal{T}_{\mathrm{stab}}$). Unit: dimensionless. Status: ASSUMED.
- S_{Ψ}: The action of dirac16complex in the bulk (kinetic term with the canonical spin connection, mass term, contact self-interaction). Unit: dimensionless. Status: RECORD.
- S_{fold}: The action of the fold: the Standard Model, the portals of dirac16complex, the fold fluid and the induced gravity of the fold, all at $\sigma = 0$. Unit: dimensionless. Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\, \mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\, \mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\, \mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- δ_{D}: The Dirac delta function: $\int \mathrm{d}\sigma\, \delta_{\mathrm{D}}(\sigma) F(\sigma) = F(0)$. Unit: 1/length (for $\delta_{\mathrm{D}}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\, \mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z) \ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4} b_4)^3$ at a time when $\mathrm{e}^{-a_4} b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathcal{L}_{\mathrm{IG}}$: The Lagrangian of the induced gravity of the fold. Unit: GeV^4. Status: ASSUMED.
- $\mathcal{L}_{\mathrm{ff}}$: The Lagrangian of the fold fluid (only its energy-momentum tensor $\mathcal{T}_{\mathrm{ff}}$ enters the equations). Unit: GeV^4. Status: ASSUMED.
- $\mathcal{L}_{\mathrm{H}}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\mathcal{L}_{\mathrm{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{f}}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{Y}}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{P}}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The action has four parts: Einstein-Lovelock gravity in eight dimensions (S_{EL}, IN-2, IN-3), the stabiliser of the bulk (S_{stab}, IN-7), the field dirac16complex in the bulk (S_{Ψ}, IN-4 to IN-6) and the fold (S_{fold}, IN-8 to IN-16). The Dirac delta $\delta_{\mathrm{D}}(\sigma)$ of the proper hidden coordinate σ confines the terms of the fold to $\sigma = 0$; the factor $1/V_*$ turns the integral over the extra times into the normalisation of four-dimensional fields (IN-8). On the fold live: the induced

gravity of the fold ($\mathcal{L}_{\rm IG}$), the fold fluid ($\mathcal{L}_{\rm ff}$), the Higgs doublet ($\mathcal{L}_{\rm H}$), the gauge fields ($\mathcal{L}_{\rm gauge}$), the quarks and leptons ($\mathcal{L}_{\rm f}$), their Yukawa couplings ($\mathcal{L}_{\rm Y}$) and the portals of dirac16complex ($\mathcal{L}_{\rm P}$). Varying $S_{\rm tot}$ with respect to the metric gives the gravitational equations (FE-1, FE-4); with respect to $\bar{\Psi}$ the equation of dirac16complex (FE-12, FE-13); with respect to $\Phi^\dagger$ the Higgs equation (FE-15); with respect to the gauge fields the Yang-Mills equations (FE-16 to FE-18); with respect to every fermion its Dirac equation (FE-19 to FE-24). Status: ASSUMED (the choice and the place of the terms; the content of each term is stated with its own label).

2.2 Gravity

$$S_{\rm EL} = \frac{1}{2\kappa}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{L}_{(p)} - 2\Lambda\Big), \qquad \kappa = 8\pi G_8,$$
$$\alpha_{(p)} = w_{(p)}\,H^{2-2p}, \qquad w_{(1)}=1, \quad w_{(2)}=\frac{1}{32}, \quad w_{(3)}=\frac{1}{1152}, \qquad \Lambda = -10H^2 \tag{IN-2}$$

Symbols in (IN-2).

- $S_{\rm EL}$: The Einstein-Lovelock action of gravity in eight dimensions. Unit: dimensionless. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- d: The differential: dX is an infinitesimal change of X, dX/dY the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0,\dots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \dots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\dots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: 1/length^{2p}. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$ (the author writes the pure number $\Lambda_{\rm N} = 10$ in $0 = -\Lambda_{\rm N} + \sum_p H^{-2p}w_{(p)}\mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_8: The eight-dimensional Newton constant ($\kappa = 8\pi G_8$). Unit: GeV^{-6}. Status: AUTHOR.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]). Unit: dimensionless. Value: $(1, 1/32, 1/1152)$. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

Gravity is the Einstein-Lovelock theory in eight dimensions with the author's weights. $\mathcal{L}_{(1)}$ is the Einstein-Hilbert density (the Ricci scalar), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density and $\mathcal{L}_{(3)}$ the cubic Lovelock density; in eight dimensions the series ends with $p = 3$ (the density of order 4 is a total derivative in eight dimensions). The author writes the vacuum equations as $0 = -\Lambda_{\rm N} + H^{-2}w_{(1)}\mathcal{E}_{(1)} + H^{-4}w_{(2)}\mathcal{E}_{(2)} + H^{-6}w_{(3)}\mathcal{E}_{(3)}$ with the pure number $\Lambda_{\rm N} = 10$ (`Dirac_Einstein-Lovelock_4+4_Nash.nb`, In[134]); multiplied by H^2 this is FE-1 without sources and with $\Lambda = -\Lambda_{\rm N}H^2 = -10H^2$. Gravity couples to every field through the metric that appears in every other term of the action (in $\sqrt{|g|}$, in the contractions of indices and in the spin connection). Status: AUTHOR.

$$\mathcal{L}_{(p)} = \frac{1}{2^p}\,\delta^{M_1N_1\cdots M_pN_p}_{P_1Q_1\cdots P_pQ_p}\,R^{P_1Q_1}{}_{M_1N_1}\cdots R^{P_pQ_p}{}_{M_pN_p},$$
$$\mathcal{L}_{(1)} = R, \qquad \mathcal{L}_{(2)} = R^2 - 4R_{MN}R^{MN} + R_{MNPQ}R^{MNPQ} \tag{IN-3}$$

Symbols in (IN-3).

- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: 1/length^{2p}. Status: STANDARD.

- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor (convention $[\nabla_P, \nabla_Q]V^M = R^M{}_{NPQ}V^N$), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: 1/length^2. Status: STANDARD.

The Lovelock densities are built with the generalised Kronecker delta from p Riemann tensors. Their key property (Lovelock's theorem, STANDARD) is that their metric variation contains no derivative of the metric higher than the second, so that the field equations FE-1 are second-order differential equations, and that it is linear in the second derivatives along any one direction (quasi-linearity): a thin source at the fold produces a finite jump of first derivatives, the junction conditions FE-4. Status: STANDARD.

2.3 The field dirac16complex

$$S_\Psi = \int \mathrm{d}^8X\, \sqrt{|g|}\, \Big[\frac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m_\Psi\, \bar\Psi\Psi - \frac{\lambda_\Psi}{2}\big(\bar\Psi\Psi\big)^2\Big] \tag{IN-4}$$

Symbols in (IN-4).

- S_Ψ: The action of dirac16complex in the bulk (kinetic term with the canonical spin connection, mass term, contact self-interaction). Unit: dimensionless. Status: RECORD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: GeV$^{7/2}$. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.

dirac16complex Ψ is the author's complex spinor field of eight dimensions with 16 components. Its Lagrangian is that of the record: the kinetic term with the canonical spin connection (in the symmetric form, which is real without a factor i for the author's real matrices, IN-5), the mass term and a contact self-interaction. It couples to gravity

through $\sqrt{|g|}$, γ^M and the spin connection Ω_M inside D_M (IN-6), and to itself through λ_Ψ. $\bar\Psi\Psi = \bar\Psi_{\rm L}\Psi_{\rm L} + \bar\Psi_{\rm R}\Psi_{\rm R}$ (the mass term does not mix the halves, IN-5), while the kinetic term mixes them (it contains one Dirac matrix). Status: RECORD.

$$\begin{aligned}
&\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}, \qquad \eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1), \qquad T^a = \eta^{ab}T_b, \qquad \gamma^M = \mathbf{e}_a{}^M\, T^a,\\
&\bar\Psi = \Psi^\dagger\sigma_{16}, \qquad \sigma_{16} = T_0T_1T_2T_3, \qquad \Gamma_8 = T_0T_1T_2T_3T_4T_5T_6T_7,\\
&P_{\rm L} = \frac{\mathbf{1}-\Gamma_8}{2}, \qquad P_{\rm R} = \frac{\mathbf{1}+\Gamma_8}{2}, \qquad \Psi_{\rm L} = P_{\rm L}\Psi, \qquad \Psi_{\rm R} = P_{\rm R}\Psi
\end{aligned} \tag{IN-5}$$

Symbols in (IN-5).

- T_a, T^a: The author's eight real 16×16 Dirac matrices T16ˆA[a] of 4+4.nb ($a = 0,\dots,7$; $T_0,\dots,T_7$), with $\{T_a,T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.
- a: A frame (orthonormal) index running over $0,1,\dots,7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a,b\in\{1,\dots,7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$ (eta4488 of 4+4.nb); η^{ab} is its inverse (the same matrix). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \dots, x_7$ for a tensor of eight dimensions, $x_1,\dots,x_7$ for a tensor of the fold, $x_1,\dots,x_4$ for a four-dimensional tensor). Status: STANDARD.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \dots$): it runs over $x_0, x_1, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16} = T_0T_1T_2T_3$, real and symmetric; $\sigma_{16}T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.
- Γ_8: The chirality matrix of eight dimensions, $\Gamma_8 = T_0T_1\cdots T_7$ (the author's T16ˆA[8]) $= \mathrm{diag}(-\mathbf{1}_8, \mathbf{1}_8)$; $\Gamma_8^2 = \mathbf{1}$; it anticommutes with every T_a. Unit: dimensionless. Status: AUTHOR.
- $P_{\rm L}$: The author's left-handed chiral projector, $P_{\rm L} = (\mathbf{1}-\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- $P_{\rm R}$: The author's right-handed chiral projector, $P_{\rm R} = (\mathbf{1}+\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- $\Psi_{\rm L}$: The left-handed half of dirac16complex, $\Psi_{\rm L} = P_{\rm L}\Psi$ (the upper eight components in the basis of 4+4.nb). Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- $\Psi_{\rm R}$: The right-handed half of dirac16complex, $\Psi_{\rm R} = P_{\rm R}\Psi$ (the lower eight components). Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.

T_a ($a = 0,\dots,7$) are the author's eight real 16×16 Dirac matrices `T16ˆA[a]` of `4+4.nb`, and Γ_8 is the author's chirality matrix `T16ˆA[8]` $= \mathrm{diag}(-\mathbf{1}_8,\mathbf{1}_8)$; $P_{\rm L}$ and $P_{\rm R}$ are the author's chiral projectors (the provenance document `dirac matrices.md` of UM-3 proves the Clifford relations, the reality and the projectors; the facts used here are restated in this paragraph). The Dirac adjoint uses $\sigma_{16} = T_0T_1T_2T_3$: σ_{16} is real and symmetric, $\sigma_{16}T_a$ is real and antisymmetric, and σ_{16} commutes with Γ_8 (it is a product of four Dirac matrices), so $\overline{P_X\Psi} = \bar\Psi P_X$ and every bilinear with an even number of Dirac matrices joins a half with itself, every bilinear with an odd number joins $\Psi_{\rm L}$ with $\Psi_{\rm R}$ (DERIVED, `um3_fields.py`, checks F1 and F2). Status: AUTHOR (matrices, projectors), RECORD (adjoint).

$$\begin{aligned}
&D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \qquad D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\,\Omega_M, \qquad \Omega_M = \frac{1}{2}\,\omega_{Mab}\,\Sigma^{ab}, \qquad \Sigma^{ab} = \frac{1}{4}\big[T^a, T^b\big],\\
&\omega_{Mab} = \mathbf{e}_a{}^N\big(\partial_M\mathbf{e}_{bN} - \Gamma^P{}_{MN}\,\mathbf{e}_{bP}\big)
\end{aligned} \tag{IN-6}$$

Symbols in (IN-6).

- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- Ω_M: The spin connection acting on dirac16complex, $\Omega_M = \frac12\omega_{Mab}\Sigma^{ab}$. Unit: 1/length. Status: RECORD.
- ω_{Mab}: The canonical (Levi-Civita) spin connection of the vielbein, antisymmetric in a, b. Unit: 1/length. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac14[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu\mathbf{e}_b{}^\nu\Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- T_a, T^a: The author's eight real 16×16 Dirac matrices T16ˆA[a] of 4+4.nb ($a = 0, \ldots, 7$; $T_0, \ldots, T_7$), with $\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Gamma^P{}_{MN}$: The Christoffel symbols of the metric g_{MN} (the Levi-Civita connection). Unit: 1/length. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

The canonical spin connection ω_{Mab} of the vielbein (antisymmetric in a, b) acts on the spinor through the generators Σ^{ab} of Spin(4,4). With it the Dirac matrices are covariantly constant ($D_M\gamma^N = 0$), the kinetic term of IN-4 is a scalar and the field equation FE-12 is covariant. For the author's metric $\gamma^M\Omega_M$ is never zero (FE-14). Status: RECORD (the canonical construction), DERIVED (the vielbein postulate for the author's metric, `um3_geometry.py`, G5).

2.4 The stabiliser of the bulk

$$\mathcal{T}_{\mathrm{stab}}{}^a{}_b = \mathrm{diag}\big(p_\mathrm{s}, p_\mathrm{s}, p_\mathrm{s}, p_\mathrm{s}, -\rho_\mathrm{s}, p_\mathrm{s}, p_\mathrm{s}, p_\mathrm{s}\big), \qquad \kappa\rho_\mathrm{s} = -\frac{H^2}{16}, \qquad \kappa p_\mathrm{s} = -\frac{5H^2}{16} \tag{IN-7}$$

Symbols in (IN-7).

- $\mathcal{T}_{\mathrm{stab}}{}^a{}_b$: The energy-momentum tensor of the stabiliser of the bulk (frame components). Unit: GeV^8. Status: ASSUMED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- p_s: The pressure of the stabiliser (equal along σ, ordinary space and the extra times). Unit: GeV^8. Value: $\kappa p_\mathrm{s} = -5H^2/16$. Status: ASSUMED.
- ρ_s: The energy density of the stabiliser. Unit: GeV^8. Value: $\kappa\rho_\mathrm{s} = -H^2/16$. Status: ASSUMED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

$S_{\rm stab}$ is the action of a medium of the bulk; only its energy-momentum tensor $\mathcal{T}_{\rm stab}$ enters the field equations, and IN-7 states it in the frame of the author's metric. With the author's weights and Λ, the bulk of a universe whose ordinary space expands slowly (the late universe, $a_4' \ll H$) is not a vacuum solution: its frame curvature is that of the static block (FE-32), which requires exactly this source, a negative energy density $\rho_{\rm s} = -H^2/(16\kappa)$ with the isotropic pressure $p_{\rm s} = 5\rho_{\rm s}$ (DERIVED, `um3_geometry.py`, G4). The stabiliser couples only to gravity. It is absent in the primordial era, whose bulk is the author's vacuum (FE-31). Status: ASSUMED (its existence; its value is DERIVED from the requirement of a static bulk); its microscopic origin is OPEN.

2.5 The fold

$$\int \mathrm{d}^8X\,\sqrt{|g|}\,\delta_{\rm D}(\sigma)\,\frac{1}{V_*}\,F = \int \mathrm{d}^4x\,\sqrt{-\hat{h}}\,\mathcal{Z}\,F\big|_{\sigma=0}, \qquad \mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4,$$
$$\hat{h}_{\mu\nu} = \mathrm{diag}\big(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1\big), \qquad \sqrt{-\hat{h}} = \mathrm{e}^{3a_4}, \qquad \mathcal{Z} = \frac{V_{\rm t}\big(\mathrm{e}^{-a_4}b_4\big)^3}{V_*}, \qquad V_{\rm t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7 \tag{IN-8}$$

Symbols in (IN-8).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- $\delta_{\rm D}$: The Dirac delta function: $\int \mathrm{d}\sigma\,\delta_{\rm D}(\sigma)F(\sigma) = F(0)$. Unit: 1/length (for $\delta_{\rm D}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\rm t}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- F: Any function of the fields at the fold (used to state a rule of integration). Status: STANDARD.
- x: The coordinates of the fold: x_i ($i = 1, 2, 3$) of ordinary space, x_4 the time, x_r ($r = 5, 6, 7$) the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_{\rm t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in

the late universe. Status: DERIVED.
- V_t: The coordinate volume of the (compact) extra times, $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$. Unit: length^3. Status: ASSUMED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The fold is the seven-dimensional surface $\sigma = 0$ ($z = \pi/2$) with the coordinates $x_1, \ldots, x_7$. No field depends on the extra times (ASSUMED, FE-38), so the integral over them gives their coordinate volume V_t (the extra times are compact) times their scale factors $(\mathrm{e}^{-a_4}b_4)^3$ at the fold. V_* is the physical volume of the extra times on the fold in the late universe, where $\mathrm{e}^{-a_4}b_4$ is constant (FE-33); with this normalisation $\mathcal{Z} = 1$ in the whole late universe, the fields of the fold are four-dimensional fields in the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold, and their couplings are the measured four-dimensional ones. $\mathcal{Z}(x_4)$ is the factor by which the extra times rescale every coupling of the fold when $\mathrm{e}^{-a_4}b_4$ changes (FE-44). F stands for any function of the fields on the fold. Status: ASSUMED (the place of the fields), DERIVED (the reduction).

$$\mathcal{L}_\mathrm{IG} = \frac{\hat{R}}{2\kappa_4} \tag{IN-9}$$

Symbols in (IN-9).

- $\mathcal{L}_\mathrm{IG}$: The Lagrangian of the induced gravity of the fold. Unit: GeV^4. Status: ASSUMED.
- $\hat{R}$: The scalar curvature of the metric $\hat{h}_{\mu\nu}$ of the fold. Unit: 1/length^2. Status: STANDARD.
- κ_4: The coupling of the induced gravity of the fold, $\kappa_4 = 8\pi G_\mathrm{ind}$; four-dimensional gravity on the fold has $8\pi G_\mathrm{N} = \kappa_4/(1 + 5r/6)$ (FE-59). Unit: GeV^{-2}. Value: $\kappa_4 \simeq 8\pi G_\mathrm{N} = 8\pi/M_\mathrm{Pl}^2$. Status: ASSUMED.

The induced gravity of the fold: an Einstein-Hilbert term of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold ($\hat{R}$ its scalar curvature), with the coupling $\kappa_4 = 8\pi G_\mathrm{ind}$. Such a term is generated by the quantum fluctuations of every field that lives on the fold (induced gravity); UM-3 assumes that it dominates four-dimensional gravity, because the gravitons of the bulk move with the wrong speed (DERIVED, `um3_bulk.py`, B5: $c_\mathrm{g}^2 = 15/2$ at the static block). With it, four-dimensional gravity on the fold is general relativity with Newton's constant G_N (FE-59), up to a fraction r of bulk gravity. Status: ASSUMED.

$$\mathcal{T}_\mathrm{ff}{}^a{}_b = \mathrm{diag}\big(p_\mathrm{ff}, p_\mathrm{ff}, p_\mathrm{ff}, -\rho_\mathrm{ff}, q_\mathrm{ff}, q_\mathrm{ff}, q_\mathrm{ff}\big) \tag{IN-10}$$

Symbols in (IN-10).

- $\mathcal{T}_\mathrm{ff}{}^a{}_b$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- p_ff: The pressure of the fold fluid along ordinary space. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.
- ρ_ff: The energy density of the fold fluid. Unit: GeV^4. Value: $31HV_*/(4\kappa)$ at the static block. Status: ASSUMED.
- q_ff: The pressure of the fold fluid along the extra times. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.

The fold fluid: a medium at rest on the fold (the frame indices a, b run over the seven directions $1, \ldots, 7$ of the fold), with the energy density ρ_ff, the pressure p_ff along ordinary space and the pressure q_ff along the extra times (four-dimensional normalisation, IN-8). The kink of the author's metric at the fold requires a thin source

(FE-4); the fold fluid is the part of it that the Standard Model and the portals do not supply: at the static block $\rho_{\rm ff} = 31HV_*/(4\kappa)$, $p_{\rm ff} = q_{\rm ff} = -61HV_*/(8\kappa)$ (FE-32), at the author's vacuum $\rho_{\rm ff} = 6HV_*/\kappa$, $p_{\rm ff} = q_{\rm ff} = -10HV_*/\kappa$ less the contribution of the induced gravity (FE-31). It plays the role of the brane tension of brane-world models. $\mathcal{L}_{\rm ff}$ stands for its Lagrangian; only $\mathcal{T}_{\rm ff}$ enters the field equations. Status: ASSUMED (its existence; its values DERIVED from the junction conditions); its microscopic origin is OPEN.

2.6 The Higgs doublet with the ad hoc vacuum

$$\mathcal{L}_{\rm H} = -\hat{h}^{\mu\nu}\big(D_\mu\Phi\big)^\dagger D_\nu\Phi - V(\Phi), \qquad V(\Phi) = \mu^2\,\Phi^\dagger\Phi + \lambda\big(\Phi^\dagger\Phi\big)^2, \qquad \lambda > 0, \quad \mu^2 < 0 \tag{IN-11}$$

Symbols in (IN-11).

- $\mathcal{L}_{\rm H}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $V(\Phi)$: The Higgs potential, $V = \mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2$. Unit: GeV^4. Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.

The Higgs doublet Φ (two complex components, hypercharge 1/2) with the author's potential. In the signature $(+,+,+,-)$ of the fold the kinetic term carries a minus sign, so that the energy density is positive. Status: AUTHOR (the potential), STANDARD (the rest).

$$\Phi = \frac{1}{\sqrt{2}}\begin{pmatrix}\mathcal{A}(x_4)\\ v+h\end{pmatrix}, \qquad \mathcal{A}(x_4) = v_{\mathcal{A}}\,\mathrm{e}^{-2a_{40}x_4}, \qquad a_{40} > 0, \quad x_4 \geq 0, \qquad v^2 = -\frac{\mu^2}{\lambda} \tag{IN-12}$$

Symbols in (IN-12).

- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_{\mathcal{A}}\mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV. Status: STANDARD.
- h: The Higgs field: the fluctuation of the lower component of the Higgs doublet about v. Unit: GeV. Value: $m_h = 125.25$ GeV. Status: STANDARD.
- $v_{\mathcal{A}}$: The amplitude of almightyAdHoc at $x_4 = 0$; the author's almightyAdHoc = Exp$[-2\,a_{40}\,x_4]$ has $v_{\mathcal{A}} = 1$ in the unit of the field. Unit: GeV. Value: 1 (author). Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_{40}: The decay constant of almightyAdHoc, $a_{40} > 0$: $\mathcal{A}$ falls by e in the time $1/(2a_{40})$. Unit: GeV (1/length). Status: AUTHOR.

- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\text{ GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.

The author's ad hoc vacuum: the upper component of the Higgs doublet is the prescribed function almightyAdHoc $\mathcal{A}(x_4) = \exp(-2a_{40}x_4)$ (the author writes it with the amplitude $v_\mathcal{A} = 1$ in the unit of the field; UM-3 keeps $v_\mathcal{A}$ to show where it enters), the lower component is the vacuum expectation value v plus the Higgs field h. The Goldstone fields are absorbed by the gauge choice (unitary gauge). $\mathcal{A}$ decays with the rate $2a_{40}$ from the beginning of the time $x_4 = 0$: it acts only in the primordial era. The ad hoc vacuum is not an exact solution of the Higgs equation; the residuals are FE-26. Status: AUTHOR.

2.7 The gauge fields

$$
\begin{aligned}
D_\mu &= \partial_\mu + \hat{\omega}_\mu - \mathrm{i}g_3\, G_\mu^A\, t^A - \mathrm{i}g_2\, W_\mu^I\, \frac{\tau^I}{2} - \mathrm{i}g_1\, Y_f\, B_\mu, \\
\mathcal{L}_{\text{gauge}} &= -\frac{1}{4}\, G^A_{\mu\nu} G^{A\mu\nu} - \frac{1}{4}\, W^I_{\mu\nu} W^{I\mu\nu} - \frac{1}{4}\, B_{\mu\nu} B^{\mu\nu}, \\
G^A_{\mu\nu} &= \partial_\mu G^A_\nu - \partial_\nu G^A_\mu + g_3\, f^{ABC} G^B_\mu G^C_\nu, \qquad W^I_{\mu\nu} = \partial_\mu W^I_\nu - \partial_\nu W^I_\mu + g_2\, \epsilon^{IJK} W^J_\mu W^K_\nu, \qquad B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu
\end{aligned}
\tag{IN-13}
$$

Symbols in (IN-13).

- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat{h}^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- $\hat{\omega}_\mu$: The spin connection of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold acting on four-dimensional spinors. Unit: 1/length. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling (the strong coupling g_s). Unit: dimensionless. Value: $g_3(m_Z) = 1.2177$ ($\alpha_s = 0.1180$). Status: STANDARD.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- B: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- C: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- t^A: The generators of $SU(3)$ on colour triplets, $t^A = \lambda^A_{\text{GM}}/2$ (the Gell-Mann matrices halved), $\mathrm{tr}(t^A t^B) = \delta^{AB}/2$. Unit: dimensionless. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- J: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- K: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- g_1: The $U(1)_Y$ gauge coupling (usually g'; not normalised for grand unification). Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- Y_f: The hypercharge of a field (IN-13: 1/6, 2/3, $-1/3$, $-1/2$, -1, 0 for $Q, u_\text{R}, d_\text{R}, \ell, e_\text{R}, \nu_\text{R}$, 1/2 for Φ). Unit: dimensionless. Status: STANDARD.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over

the six. Status: STANDARD.

- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- $\mathcal{L}_{\text{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- f^{ABC}: The structure constants of $SU(3)$, $[t^A, t^B] = \mathrm{i} f^{ABC} t^C$. Unit: dimensionless. Status: STANDARD.
- ϵ^{IJK}: The Levi-Civita symbol of three indices (the structure constants of $SU(2)$), $\epsilon^{123} = 1$. Unit: dimensionless. Status: STANDARD.

The gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold, with the canonical normalisation (the couplings in the covariant derivative): the gluons G^A_μ, the weak-isospin fields $W^1_\mu, W^2_\mu, W^3_\mu$ and the hypercharge field B_μ. The covariant derivative of a field acts with the part that fits the field: t^A on colour triplets, $\tau^I/2$ on weak doublets, the hypercharge Y_f of the field, and the four-dimensional spin connection $\hat{\omega}_\mu$ on spinors. The hypercharges (STANDARD) are $Y = 1/6$ for Q, $2/3$ for u_R, $-1/3$ for d_R, $-1/2$ for ℓ, -1 for e_R, 0 for ν_R and $1/2$ for Φ. The gauge fields have no components along the extra times (ASSUMED: such components would be scalars of wrong sign). Status: STANDARD.

2.8 The quarks and leptons

$$\mathcal{L}_\mathrm{f} = \sum_\psi \mathrm{i}\,\bar{\psi}\,\hat{\gamma}^\mu D_\mu \psi, \qquad \psi \in \{Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}\},$$
$$\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}, \qquad \bar{\psi} = \psi^\dagger\beta, \qquad \beta = \hat{\gamma}^{(4)}, \qquad \hat{\gamma}_5 = \mathrm{i}\,\hat{\gamma}^{(4)}\hat{\gamma}^{(1)}\hat{\gamma}^{(2)}\hat{\gamma}^{(3)},$$
$$\hat{P}_\mathrm{L} = \frac{\mathbf{1} - \hat{\gamma}_5}{2}, \quad \hat{P}_\mathrm{R} = \frac{\mathbf{1} + \hat{\gamma}_5}{2}, \qquad \hat{P}_\mathrm{L} Q = Q, \quad \hat{P}_\mathrm{L}\ell = \ell, \quad \hat{P}_\mathrm{R} u_\mathrm{R} = u_\mathrm{R}, \quad \hat{P}_\mathrm{R} d_\mathrm{R} = d_\mathrm{R}, \quad \hat{P}_\mathrm{R} e_\mathrm{R} = e_\mathrm{R}, \quad \hat{P}_\mathrm{R}\nu_\mathrm{R} = \nu_\mathrm{R} \tag{IN-14}$$

Symbols in (IN-14).

- $\mathcal{L}_\mathrm{f}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ψ, $\bar{\psi}$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar{\psi} = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu \hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- u_R: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_R: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the

adjoint operator). Status: STANDARD.
- β: The matrix of the four-dimensional Dirac adjoint, $\beta = \hat{\gamma}^{(4)}$ (Hermitian, $\beta^2 = \mathbf{1}$). Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}_5$: The four-dimensional chirality matrix, $\hat{\gamma}_5 = \mathrm{i}\hat{\gamma}^{(4)}\hat{\gamma}^{(1)}\hat{\gamma}^{(2)}\hat{\gamma}^{(3)}$, $\hat{\gamma}_5^2 = \mathbf{1}$. Unit: dimensionless. Status: STANDARD.
- $\hat{P}_\mathrm{L}$: The four-dimensional left-handed projector, $\hat{P}_\mathrm{L} = (\mathbf{1} - \hat{\gamma}_5)/2$. Unit: dimensionless. Status: STANDARD.
- $\hat{P}_\mathrm{R}$: The four-dimensional right-handed projector, $\hat{P}_\mathrm{R} = (\mathbf{1} + \hat{\gamma}_5)/2$. Unit: dimensionless. Status: STANDARD.

Every Standard-Model fermion is a four-dimensional Weyl field on the fold, written as a four-component spinor with a fixed chirality: the quark doublets Q and the lepton doublets ℓ are left-handed, the singlets u_R, d_R, e_R, ν_R right-handed. Each symbol stands for the three generations (a vector in generation space) and, for quarks, the three colours. $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu \hat{\gamma}^{(a)}$ are complex 4×4 Dirac matrices of the fold (with the frame matrices $\hat{\gamma}^{(a)}$, $a = 1, \ldots, 4$, $(\hat{\gamma}^{(4)})^2 = \mathbf{1}$, $(\hat{\gamma}^{(1)})^2 = (\hat{\gamma}^{(2)})^2 = (\hat{\gamma}^{(3)})^2 = -\mathbf{1}$); $\beta = \hat{\gamma}^{(4)}$ is Hermitian and $(\hat{\gamma}^\mu)^\dagger = \beta\hat{\gamma}^\mu\beta$. With these conventions the free Dirac equation is $(-\mathrm{i}\hat{\gamma}^\mu\partial_\mu + m)\psi = 0$ and $\bar{\psi}\beta\psi$-type energy densities are positive. Status: STANDARD.

$$\mathcal{L}_\mathrm{Y} = -\big(\bar{Q}\,\mathcal{Y}^u\,\tilde{\Phi}\,u_\mathrm{R} + \bar{Q}\,\mathcal{Y}^d\,\Phi\,d_\mathrm{R} + \bar{\ell}\,\mathcal{Y}^e\,\Phi\,e_\mathrm{R} + \bar{\ell}\,\mathcal{Y}^\nu\,\tilde{\Phi}\,\nu_\mathrm{R}\big) + \mathrm{h.c.}, \qquad \tilde{\Phi} = \mathrm{i}\tau^2\Phi^* \tag{IN-15}$$

Symbols in (IN-15).

- $\mathcal{L}_\mathrm{Y}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge $1/6$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- u_R: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge $2/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge $1/2$), a field of the fold. Unit: GeV. Status: AUTHOR.
- d_R: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- h.c.: The Hermitian conjugate of the preceding terms. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- *: Complex conjugation of a number, or of each entry of a matrix or column. Status: STANDARD.

The Yukawa couplings, with the complex 3×3 matrices $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$ in generation space; $\tilde{\Phi}$ is the conjugate doublet (hypercharge $-1/2$). In the vacuum they give the quarks, the charged leptons and the neutrinos their Dirac masses (FE-46); with three right-handed neutrinos the neutrino masses are Dirac masses and the lepton mixing is that of the PMNS matrix. No Majorana mass of ν_R is included (lepton number is conserved). Status: STANDARD (with ν_R: accepted by the author, 2026-10-24 17:32).

2.9 The portals of dirac16complex

$$\begin{aligned}\mathcal{L}_\mathrm{P} = &-c_\Phi\,\big(\Phi^\dagger\Phi\big)\,\bar{\Psi}P_\mathrm{L}\Psi - \sum_X \Big[\frac{1}{4}\big(c_X^G\,G^A_{\mu\nu}G^{A\mu\nu} + c_X^W\,W^I_{\mu\nu}W^{I\mu\nu} + c_X^B\,B_{\mu\nu}B^{\mu\nu}\big)\,\bar{\Psi}P_X\Psi \\ &+ \frac{d_X}{2}\,B_{\mu\nu}\,\mathcal{D}_X^{\mu\nu} + \sum_f c_X^f\,J_f^\mu\,j_{X\mu}\Big], \\ \mathcal{D}_X^{\mu\nu} = \mathrm{i}\,\bar{\Psi}\,\Sigma^{\mu\nu}P_X\Psi, &\qquad j_X^\mu = \mathrm{i}\big(\bar{\Psi}P_X D^\mu\Psi - (D^\mu\bar{\Psi})P_X\Psi\big), \qquad J_f^\mu = \bar{\psi}_f\,\hat{\gamma}^\mu\psi_f\end{aligned} \tag{IN-16}$$

Symbols in (IN-16).

- $\mathcal{L}_\mathrm{P}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G_μ^A, $G_{\mu\nu}^A$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W_μ^I, $W_{\mu\nu}^I$: The gauge field of $SU(2)_L$ on the fold, $W_\mu^1, W_\mu^2, W_\mu^3$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar\Psi\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind). Unit: GeV^{-7}. Status: INVENTED.
- J_f^μ: The fermion-number current of the Standard-Model multiplets of kind f, $J_f^\mu = \bar\psi_f\hat\gamma^\mu\psi_f$ summed over the three generations and the gauge components; with other labels the gauge currents (FE-16 to FE-18). Unit: GeV^3. Status: STANDARD.
- j_X^μ: The convective current of the half X of dirac16complex, $j_X^\mu = \mathrm{i}(\bar\Psi P_X D^\mu\Psi - (D^\mu\bar\Psi)P_X\Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu\mathbf{e}_b{}^\nu\Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- ψ, $\bar\psi$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar\psi = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat\gamma^\mu = \hat{\mathbf{e}}_a{}^\mu\hat\gamma^{(a)}$, $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.

The new interactions of UM-3 (INVENTED), the order of the author that the left-handed and the right-handed halves of dirac16complex each couple to every Standard-Model field with its own coupling, and that only the left-handed half couples to the Higgs doublet. Ψ is the value of dirac16complex at the fold. Every term is a product of a gauge-invariant Standard-Model operator and a Lorentz-tensor bilinear of one half X of dirac16complex, so

every term is invariant under $SU(3)_C \times SU(2)_L \times U(1)_Y$ and under the Lorentz transformations of the fold; none changes the flavour of a quark or a lepton (consistent with QCD and with the electroweak theory, DERIVED by construction):

- the Higgs portal: the scalar density $\bar{\Psi} P_{\mathrm{L}} \Psi$ of the left-handed half times $\Phi^\dagger \Phi$, with the coupling c_Φ; the right-handed half has no Higgs coupling;
- the gauge-kinetic portals: the scalar density of each half times the gauge-invariant kinetic densities of the gluons, of the W fields and of the B field, with the six couplings c_X^G, c_X^W, c_X^B: they make the gauge couplings of the fold depend on the density of dirac16complex at the fold (FE-16 to FE-18) and couple the bound states of dirac16complex to gluons, photons and weak bosons;
- the dipole portal: the antisymmetric tensor $\mathcal{D}_X^{\mu\nu}$ of each half times the hypercharge field strength (the only Standard-Model field strength that is gauge invariant by itself), with the couplings d_X: an electric and magnetic dipole moment of dirac16complex with respect to the photon and the Z;
- the current portals: the convective current j_X^μ of each half (the only real vector current that joins a half with itself: $\bar{\Psi} \gamma^\mu P_X \Psi$ joins Ψ_{L} with Ψ_{R} and is not real, DERIVED, `um3_fields.py`, F1) times the fermion-number current J_f^μ of each kind f of Standard-Model multiplet, with the twelve couplings c_X^f; J_f^μ sums over the three generations and over the gauge components, so the portal is flavour blind.

The 21 couplings are c_Φ, c_{L}^G, c_{R}^G, c_{L}^W, c_{R}^W, c_{L}^B, c_{R}^B, d_{L}, d_{R} and c_{L}^f, c_{R}^f for $f = Q, u, d, \ell, e, \nu$. Every term is bilinear in Ψ: the action is invariant under $\Psi \to \mathrm{e}^{\mathrm{i}\theta} \Psi$ (the charge of dirac16complex is conserved, and its lightest state is stable), and baryon and lepton numbers are conserved. Status: INVENTED.

3 Gravity

3.1 The Einstein-Lovelock equations in the bulk

$$\sum_{p=1}^{3} \alpha_{(p)} \, {\mathcal{E}_{(p)}}^M{}_N + \Lambda \, \delta^M_N = \kappa \big({\mathcal{T}_\Psi}^M{}_N + {\mathcal{T}_{\mathrm{stab}}}^M{}_N \big) \qquad (\sigma \neq 0) \tag{FE-1}$$

Symbols in (FE-1).

- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^M{}_N$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$ (the author writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex, $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\, \delta S_\Psi / \delta g^{MN}$. Unit: GeV^8. Status: STANDARD.
- ${\mathcal{T}_{\mathrm{stab}}}^a{}_b$: The energy-momentum tensor of the stabiliser of the bulk (frame components). Unit: GeV^8. Status: ASSUMED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z| \, \mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z) \ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The variation of S_{tot} with respect to the metric, away from the fold. The left-hand side is the Lovelock tensor of the author's theory; the right-hand side is κ times the energy-momentum tensor of the fields of the bulk: dirac16complex

(FE-6) and the stabiliser (IN-7). With the energy-momentum tensor defined by $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\,\delta S_{\text{matter}}/\delta g^{MN}$, the time-time component $\mathcal{T}_{x_4x_4} = -\mathcal{T}^{x_4}{}_{x_4}$ is the energy density seen by an observer at rest. Status: AUTHOR (the theory), STANDARD (the variation).

$$
\begin{aligned}
\mathcal{E}_{(p)}{}^{M}{}_{N} &= -\frac{1}{2^{p+1}}\,\delta^{MM_1N_1\cdots M_pN_p}_{NP_1Q_1\cdots P_pQ_p}\,R^{P_1Q_1}{}_{M_1N_1}\cdots R^{P_pQ_p}{}_{M_pN_p},\\
\mathcal{E}_{(1)}{}^{M}{}_{N} &= R^{M}{}_{N} - \frac{1}{2}\,\delta^{M}_{N}\,R,\\
\mathcal{E}_{(2)}{}^{M}{}_{N} &= 2\big(R\,R^{M}{}_{N} - 2R^{M}{}_{P}R^{P}{}_{N} - 2R^{MP}{}_{NQ}R^{Q}{}_{P} + R^{MPQS}R_{NPQS}\big) - \frac{1}{2}\,\delta^{M}_{N}\,\mathcal{L}_{(2)}
\end{aligned}
\tag{FE-2}
$$

Symbols in (FE-2).

- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^{M}_{N} = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor (convention $[\nabla_P, \nabla_Q]V^M = R^{M}{}_{NPQ}V^N$), with two indices the Ricci tensor $R_{MN} = R^{P}{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\text{length}^2$. Status: STANDARD.
- S: A coordinate index of eight dimensions, running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\text{length}^{2p}$. Status: STANDARD.

The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric. For $p = 1$ it is the Einstein tensor, for $p = 2$ the Gauss-Bonnet tensor (written out in the third line); the cubic tensor $\mathcal{E}_{(3)}$ is the first line with $p = 3$. UM-3 evaluates the first line exactly by enumerating the non-zero frame components of the curvature (`um3_lovelock.py`); the second and third lines are independent checks of it (DERIVED, `um3_geometry.py`, G3: the Einstein tensor from the coordinate Ricci tensor and the explicit Gauss-Bonnet tensor at random points agree with the first line). Status: STANDARD.

$$
\mathcal{E}_{(p)}{}^{M}{}_{M} = -\frac{8-2p}{2}\,\mathcal{L}_{(p)}, \qquad \nabla_M\,\mathcal{E}_{(p)}{}^{M}{}_{N} = 0
\tag{FE-3}
$$

Symbols in (FE-3).

- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols ${\Gamma^P}_{MN}$). Status: STANDARD.

Two identities of every Lovelock tensor in eight dimensions: its trace is a multiple of the Lovelock density, and its covariant divergence vanishes (the contracted Bianchi identity). The second makes the energy-momentum tensor of the sources conserved (FE-11). Status: STANDARD (DERIVED for the author's metric with general $a_4(x_4)$, $b_4(x_4)$: `um3_geometry.py`, G3).

3.2 The junction conditions at the fold

$$\begin{aligned} \kappa\, {\mathcal{S}^a}_b &= \lim_{\varepsilon\to 0}\int_{-\varepsilon}^{\varepsilon} \mathrm{d}\sigma \sum_{p=1}^{3} \alpha_{(p)}\, {\mathcal{E}_{(p)}}^a{}_b = {\mathfrak{F}^a}_b\big|_{\sigma=0^+} - {\mathfrak{F}^a}_b\big|_{\sigma=0^-}, \\ \frac{\partial \mathfrak{F}^a{}_b}{\partial(\partial_\sigma \mathfrak{f})} &= \frac{\partial}{\partial(\partial_\sigma^2 \mathfrak{f})} \sum_{p=1}^{3} \alpha_{(p)}\, {\mathcal{E}_{(p)}}^a{}_b, \qquad \mathfrak{f} \in \{\mathfrak{a}, \mathfrak{b}, \mathfrak{n}\} \end{aligned} \tag{FE-4}$$

Symbols in (FE-4).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ${\mathcal{S}^a}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- ε: The half-thickness of a layer around the fold, taken to zero. Unit: length. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^M{}_N$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- ${\mathfrak{F}^a}_b$: The junction potential: a function of the first σ-derivatives of the metric functions whose derivatives are the coefficients of the second σ-derivatives in $\sum_p \alpha_{(p)} {\mathcal{E}_{(p)}}^a{}_b$ (FE-4). Unit: GeV. Status: DERIVED.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- $\mathfrak{f}$: Any one of the three metric functions $\mathfrak{a}, \mathfrak{b}, \mathfrak{n}$ of the general warped metric (FE-34). Unit: dimensionless. Status: DERIVED.
- $\mathfrak{a}$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a} = a_4 - H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35. Unit: dimensionless. Status: DERIVED.
- $\mathfrak{b}$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$; $\mathfrak{b}_n$ its term of order n. Unit: dimensionless. Status: DERIVED.
- $\mathfrak{n}$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4 x_4} = -\mathrm{e}^{2\mathfrak{n}}$); for the author's metric $\mathfrak{n} = 0$; $\mathfrak{n}_n$ its term of order n. Unit: dimensionless. Status: DERIVED.

The field equations at the fold. The author's metric (FE-27) has a kink at $\sigma = 0$; in general (FE-34) the metric functions $\mathfrak{a}, \mathfrak{b}, \mathfrak{n}$ have first σ-derivatives that jump at the fold, so their second derivatives contain $\delta_{\mathrm{D}}(\sigma)$. Because every Lovelock tensor is linear in the second σ-derivatives (quasi-linearity, DERIVED for the general metric: `um3_fold.py`, J3), FE-1 extended across the fold reads $\sum_p \alpha_{(p)}\mathcal{E}_{(p)} + \Lambda = \kappa(\mathcal{T}_{\mathrm{bulk}} + \mathcal{S}\,\delta_{\mathrm{D}}(\sigma))$: the fold carries the thin source $\mathcal{S}^a{}_b$ (energy per unit area of the fold; a, b along the fold), equal to the integral of the Lovelock tensor across the fold. The coefficients of the second derivatives form an exact differential of the first derivatives (DERIVED, J4), with the potential $\mathfrak{F}^a{}_b$; the integral is therefore the difference of $\mathfrak{F}$ on the two sides. The (σ, σ) and (σ, x_4) components contain no second σ-derivative: they carry no thin part and hold on each side separately. Status: DERIVED.

$$\mathcal{S}^a{}_b = \frac{1}{V_*}\Big(\mathcal{T}_{\mathrm{H}}{}^a{}_b + \mathcal{T}_{\mathrm{gauge}}{}^a{}_b + \mathcal{T}_{\mathrm{f}}{}^a{}_b + \mathcal{T}_{\mathrm{P}}{}^a{}_b + \mathcal{T}_{\mathrm{ff}}{}^a{}_b + \mathcal{T}_{\mathrm{IG}}{}^a{}_b\Big), \qquad \mathcal{T}_{\mathrm{IG}}{}^\mu{}_\nu = -\frac{1}{\kappa_4}\,\hat{G}^\mu{}_\nu, \qquad \mathcal{T}^r{}_r = \mathcal{L}_{\mathrm{fold}} \tag{FE-5}$$

Symbols in (FE-5).

- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathcal{T}_{\mathrm{H}}$: The energy-momentum tensor of the Higgs doublet (four-dimensional normalisation). Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_{\mathrm{gauge}}$: The energy-momentum tensor of the gauge fields. Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_{\mathrm{f}}$: The energy-momentum tensor of the quarks and leptons. Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_{\mathrm{P}}$: The energy-momentum tensor of the portals of dirac16complex. Unit: GeV^4. Status: DERIVED.
- $\mathcal{T}_{\mathrm{ff}}{}^a{}_b$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- $\mathcal{T}_{\mathrm{IG}}$: The energy-momentum tensor of the induced gravity of the fold, $\mathcal{T}_{\mathrm{IG}}{}^\mu{}_\nu = -\hat{G}^\mu{}_\nu/\kappa_4$. Unit: GeV^4. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- κ_4: The coupling of the induced gravity of the fold, $\kappa_4 = 8\pi G_{\mathrm{ind}}$; four-dimensional gravity on the fold has $8\pi G_{\mathrm{N}} = \kappa_4/(1 + 5r/6)$ (FE-59). Unit: GeV^{-2}. Value: $\kappa_4 \simeq 8\pi G_{\mathrm{N}} = 8\pi/M_{\mathrm{Pl}}^2$. Status: ASSUMED.
- $\hat{G}^\mu{}_\nu$: The Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold. Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- $\mathcal{T}$: The energy-momentum tensor of any one term of the fold (in $\mathcal{T}^r{}_r = \mathcal{L}_{\mathrm{fold}}$). Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{fold}}$: The Lagrangian of any one term of the fold (four-dimensional normalisation). Unit: GeV^4. Status: STANDARD.

The thin source of the fold is the energy-momentum tensor of everything that lives on the fold, divided by V_* (IN-8): the Higgs doublet (FE-7), the gauge fields (FE-8), the quarks and leptons (FE-9), the portals (FE-10), the fold fluid (IN-10) and the induced gravity, whose energy-momentum tensor is $-1/\kappa_4$ times the Einstein tensor $\hat{G}^\mu{}_\nu$ of the metric of the fold. Because the fields of the fold do not depend on the extra times and have no components along them, the extra times enter only through $\sqrt{|g|}$: for every term of the fold the component along an extra time is the Lagrangian of the term, $\mathcal{T}^r{}_r = \mathcal{L}_{\mathrm{fold}}$ (DERIVED: $\delta\sqrt{|g|} = -\frac{1}{2}\sqrt{|g|}g_{rr}\delta g^{rr}$). Combining FE-4 and FE-5 gives the Einstein equations of the fold (FE-59). Status: DERIVED.

3.3 The energy-momentum tensors

$$\mathcal{T}_{\Psi\,MN} = -\frac{1}{4}\Big(\bar{\Psi}\gamma_M D_N\Psi + \bar{\Psi}\gamma_N D_M\Psi - (D_M\bar{\Psi})\gamma_N\Psi - (D_N\bar{\Psi})\gamma_M\Psi\Big) + g_{MN}\,\mathcal{L}_\Psi \tag{FE-6}$$

Symbols in (FE-6).

- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex, $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\,\delta S_\Psi/\delta g^{MN}$. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- $\mathcal{L}_\Psi$: The Lagrangian of dirac16complex, the square bracket of IN-4. Unit: GeV^8. Status: RECORD.

The symmetric energy-momentum tensor of dirac16complex, from the variation of the vielbein in IN-4 ($\mathcal{L}_\Psi$ is the Lagrangian in the square bracket of IN-4; the variation of the spin connection adds only the divergence of the spin current, which drops out of the symmetric part). On solutions of FE-12, $\mathcal{L}_\Psi = \frac{\lambda_\Psi}{2}(\bar\Psi\Psi)^2$. For a stationary state $\Psi \propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ of the free field the energy density is $\mathcal{T}_{\Psi\,x_4x_4} = -\omega\,\mathrm{i}\bar\Psi\gamma^{x_4}\Psi$: its sign is the product of the sign of the frequency and of the Krein density (FE-43). Status: STANDARD (DERIVED for the energy density).

$$\mathcal{T}_{\mathrm{H}\mu\nu} = (D_\mu\Phi)^\dagger D_\nu\Phi + (D_\nu\Phi)^\dagger D_\mu\Phi + \hat h_{\mu\nu}\,\mathcal{L}_\mathrm{H} \tag{FE-7}$$

Symbols in (FE-7).

- $\mathcal{T}_\mathrm{H}$: The energy-momentum tensor of the Higgs doublet (four-dimensional normalisation). Unit: GeV^4. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\hat h_{\mu\nu}$: The four-dimensional metric of the fold, $\hat h_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat h^{\mu\nu}$ its inverse, $\hat h$ its determinant, $\sqrt{-\hat h} = \mathrm{e}^{3a_4}$; $\hat h^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_\mathrm{H}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.

The energy-momentum tensor of the Higgs doublet (four-dimensional normalisation): its time-time component is $|D_4\Phi|^2 + \hat h^{ij}(D_i\Phi)^\dagger D_j\Phi + V \geq V$. Status: STANDARD.

$$\mathcal{T}_{\mathrm{gauge}\,\mu\nu} = G^A_{\mu\rho}G^A{}_\nu{}^\rho + W^I_{\mu\rho}W^I{}_\nu{}^\rho + B_{\mu\rho}B_\nu{}^\rho + \hat h_{\mu\nu}\,\mathcal{L}_\mathrm{gauge} \tag{FE-8}$$

Symbols in (FE-8).

- $\mathcal{T}_\mathrm{gauge}$: The energy-momentum tensor of the gauge fields. Unit: GeV^4. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.

- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \dots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \dots, 8$; a repeated index is summed. Status: STANDARD.
- ρ: A four-dimensional coordinate index (as μ). Status: STANDARD.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \dots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_{\text{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.

The energy-momentum tensor of the gauge fields; for each field its energy density is $(\vec{E}^2 + \vec{B}^2)/2$ in terms of the electric and magnetic components. Status: STANDARD.

$$\mathcal{T}_{\mathrm{f}\mu\nu} = -\frac{\mathrm{i}}{4}\sum_{\psi}\Big(\bar{\psi}\hat{\gamma}_\mu D_\nu\psi + \bar{\psi}\hat{\gamma}_\nu D_\mu\psi - (D_\mu\bar{\psi})\hat{\gamma}_\nu\psi - (D_\nu\bar{\psi})\hat{\gamma}_\mu\psi\Big) + \hat{h}_{\mu\nu}\big(\mathcal{L}_{\mathrm{f}} + \mathcal{L}_{\mathrm{Y}}\big) \tag{FE-9}$$

Symbols in (FE-9).

- $\mathcal{T}_{\mathrm{f}}$: The energy-momentum tensor of the quarks and leptons. Unit: GeV^4. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_{\mathrm{R}}, d_{\mathrm{R}}, \ell, e_{\mathrm{R}}, \nu_{\mathrm{R}}$. Status: STANDARD.
- ψ, $\bar{\psi}$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar{\psi} = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: GeV$^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \dots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_{\mathrm{f}}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\mathrm{Y}}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.

The energy-momentum tensor of the quarks and leptons (with the kinetic term in its symmetric form). For a free particle of energy E at rest in the frame of the fold its time-time component is $E\,\psi^\dagger\psi > 0$. Status: STANDARD.

$$\begin{aligned}\mathcal{T}_{\mathrm{P}\mu\nu} = \sum_X \Big[&\big(c^G_X\, G^A_{\mu\rho}G^A{}_\nu{}^\rho + c^W_X\, W^I_{\mu\rho}W^I{}_\nu{}^\rho + c^B_X\, B_{\mu\rho}B_\nu{}^\rho\big)\,\bar{\Psi}P_X\Psi \\ &+ \frac{d_X}{2}\big(B_{\mu\rho}\,\mathcal{D}_{X\nu}{}^\rho + B_{\nu\rho}\,\mathcal{D}_{X\mu}{}^\rho\big) + \sum_f \frac{c^f_X}{2}\big(J_{f\mu}\,j_{X\nu} + J_{f\nu}\,j_{X\mu}\big)\Big] + \hat{h}_{\mu\nu}\,\mathcal{L}_{\mathrm{P}}\end{aligned} \tag{FE-10}$$

Symbols in (FE-10).

- $\mathcal{T}_{\mathrm{P}}$: The energy-momentum tensor of the portals of dirac16complex. Unit: GeV^4. Status: DERIVED.

- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G_μ^A, $G_{\mu\nu}^A$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- ρ: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W_μ^I, $W_{\mu\nu}^I$: The gauge field of $SU(2)_L$ on the fold, $W_\mu^1, W_\mu^2, W_\mu^3$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar\Psi \Sigma^{\mu\nu} P_X \Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind). Unit: GeV^{-7}. Status: INVENTED.
- J_f^μ: The fermion-number current of the Standard-Model multiplets of kind f, $J_f^\mu = \bar\psi_f \hat\gamma^\mu \psi_f$ summed over the three generations and the gauge components; with other labels the gauge currents (FE-16 to FE-18). Unit: GeV^3. Status: STANDARD.
- j_X^μ: The convective current of the half X of dirac16complex, $j_X^\mu = \mathrm{i}(\bar\Psi P_X D^\mu \Psi - (D^\mu \bar\Psi) P_X \Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- $\hat h_{\mu\nu}$: The four-dimensional metric of the fold, $\hat h_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat h^{\mu\nu}$ its inverse, $\hat h$ its determinant, $\sqrt{-\hat h} = \mathrm{e}^{3a_4}$; $\hat h_{\mu\nu}^{(1)}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_\mathrm{P}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The energy-momentum tensor of the portals (from the variation of IN-16 with respect to the metric of the fold, which enters through the raised indices, through $\hat\gamma^\mu$ in J_f^μ and through $\sqrt{-\hat h}$); the Higgs portal enters only through $\hat h_{\mu\nu} \mathcal{L}_\mathrm{P}$. It is a source of the fold. Status: DERIVED.

3.4 The conservation laws

$$\begin{aligned} &\nabla_M \big(\mathcal{T}_\Psi{}^M{}_N + \mathcal{T}_\mathrm{stab}{}^M{}_N \big) = 0 \qquad (\sigma \neq 0), \\ &\hat\nabla_a \, \mathcal{S}^a{}_b = -\Big(\mathcal{T}_\mathrm{bulk}{}^{(0)}{}_b \big|_{\sigma=0^+} - \mathcal{T}_\mathrm{bulk}{}^{(0)}{}_b \big|_{\sigma=0^-} \Big) \end{aligned} \tag{FE-11}$$

Symbols in (FE-11).

- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols $\Gamma^P{}_{MN}$). Status: STANDARD.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex, $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\, \delta S_\Psi / \delta g^{MN}$. Unit: GeV^8. Status: STANDARD.
- $\mathcal{T}_\mathrm{stab}{}^a{}_b$: The energy-momentum tensor of the stabiliser of the bulk (frame components). Unit: GeV^8. Status: ASSUMED.

- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\hat{\nabla}$: The covariant derivative of the fold: with the Levi-Civita connection of the metric of the fold ($\hat{h}_{\mu\nu}$ for four-dimensional tensors, the induced metric of the seven-dimensional fold for $\hat{\nabla}_a \mathcal{S}^a{}_b$). Status: STANDARD.
- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{stab}}$; $\mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.

The energy and momentum of the bulk are conserved away from the fold (FE-3 and FE-1). At the fold the thin source exchanges energy and momentum with the bulk: its covariant divergence along the fold equals minus the jump of the flux $\mathcal{T}_{\mathrm{bulk}}{}^{(0)}{}_b$ of the bulk sources along σ (the frame component $(0), b$). For the author's metric with general $a_4(x_4)$, $b_4(x_4)$ this is an identity of the junction formula FE-30 and the bulk equation FE-29 (DERIVED, `um3_fold.py`, J6). The fluxes of dirac16complex into the bulk carry its share of this exchange. Status: DERIVED.

4 The matter fields

4.1 dirac16complex

$$\gamma^M D_M \Psi - m_\Psi \Psi - \lambda_\Psi \big(\bar{\Psi}\Psi\big)\Psi = 0 \qquad (\sigma \neq 0) \tag{FE-12}$$

Symbols in (FE-12).

- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The field equation of dirac16complex in the bulk: the variation of IN-4 with respect to $\bar{\Psi}$ (the second half of the symmetric kinetic term, integrated by parts with $D_M\gamma^M = 0$, doubles the first). With the author's real matrices the operator $\gamma^M D_M - m_\Psi$ is real: the field equation does not contain i, and a real solution exists for every complex one. Status: RECORD.

$$\Psi\big|_{\sigma=0^+} = \exp\Big(\frac{1}{V_*}\,\gamma^{(0)}\,\mathbb{M}\Big)\,\Psi\big|_{\sigma=0^-},$$
$$\mathbb{M} = c_\Phi\big(\Phi^\dagger\Phi\big)P_{\mathrm{L}} + \sum_X\Big[\frac{1}{4}\big(c_X^G\,G^A_{\mu\nu}G^{A\mu\nu} + c_X^W\,W^I_{\mu\nu}W^{I\mu\nu} + c_X^B\,B_{\mu\nu}B^{\mu\nu}\big)P_X + \frac{d_X}{2}\,B_{\mu\nu}\,\mathrm{i}\Sigma^{\mu\nu}P_X \qquad \text{(FE-13)}$$
$$+ \sum_f c_X^f\big(2\mathrm{i}\,J_f^\mu P_X D_\mu + \mathrm{i}(\hat\nabla_\mu J_f^\mu)P_X\big)\Big]$$

Symbols in (FE-13).

- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- $\mathbb{M}$: The fold operator of dirac16complex: the 16×16 matrix (with a derivative along the fold) built from the Standard-Model fields at the fold by varying the portals with respect to $\bar\Psi$ (FE-13). Unit: GeV^{-3} (so that $\mathbb{M}/V_*$ is dimensionless). Status: DERIVED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- P_{L}: The author's left-handed chiral projector, $P_{\mathrm{L}} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G_μ^A, $G_{\mu\nu}^A$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W_μ^I, $W_{\mu\nu}^I$: The gauge field of $SU(2)_L$ on the fold, $W_\mu^1, W_\mu^2, W_\mu^3$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_{\mathrm{L}}$ for $X = \mathrm{L}$, P_{R} for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu \mathbf{e}_b{}^\nu \Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.

- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind). Unit: GeV^{-7}. Status: INVENTED.
- J_f^μ: The fermion-number current of the Standard-Model multiplets of kind f, $J_f^\mu = \bar\psi_f \hat\gamma^\mu \psi_f$ summed over the three generations and the gauge components; with other labels the gauge currents (FE-16 to FE-18). Unit: GeV^3. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat h^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\hat\nabla$: The covariant derivative of the fold: with the Levi-Civita connection of the metric of the fold ($\hat h_{\mu\nu}$ for four-dimensional tensors, the induced metric of the seven-dimensional fold for $\hat\nabla_a \mathcal{S}^a{}_b$). Status: STANDARD.

The equation of dirac16complex at the fold: the variation of the portals IN-16 with respect to $\bar\Psi$ adds $-\delta_\mathrm{D}(\sigma)\,\mathbb{M}\Psi/V_*$ to FE-12. $\mathbb{M}$ is the fold operator, a 16×16 matrix (with a derivative along the fold from the current portals) built from the Standard-Model fields at the fold. Integrating FE-12 with this term across a fold of vanishing thickness gives the matching condition in the first line: the values of dirac16complex on the two sides of the fold differ by the exponential of $\gamma^{(0)}\mathbb{M}/V_*$ (the exact solution of $\partial_\sigma\Psi = \gamma^{(0)}\mathbb{M}\,\delta_\mathrm{D}(\sigma)\Psi/V_*$ in a layer in which the fold term dominates; $(\gamma^{(0)})^2 = \mathbf{1}$). The Higgs term acts on the left-handed half only. Status: DERIVED (from INVENTED terms).

$$\gamma^M\Omega_M = -3H\,\mathrm{sgn}(\sigma)\,\gamma^{(0)} + \frac{3}{2}\,\frac{b_4'}{b_4}\,\gamma^{(4)} \neq 0 \tag{FE-14}$$

Symbols in (FE-14).

- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Ω_M: The spin connection acting on dirac16complex, $\Omega_M = \frac{1}{2}\omega_{Mab}\Sigma^{ab}$. Unit: 1/length. Status: RECORD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.

The coupling of dirac16complex to gravity through the canonical spin connection, for the author's metric: the contraction $\gamma^M\Omega_M$ that enters the Dirac operator. The first term comes from the warp of the six directions $x_1, x_2, x_3, x_5, x_6, x_7$ (each contributes $\frac{1}{2}\partial_\sigma$ of the logarithm of its scale factor), the second from the time dependence of their scale factors (a_4 cancels between ordinary space and the extra times). The first term never vanishes ($H > 0$, and it only changes sign at the fold); the second is non-zero whenever b_4 depends on x_4, as the author requires. Hence the coupling of dirac16complex to gravity through the spin connection is never identically zero (DERIVED, `um3_geometry.py`, G5, with the author's matrices; the vielbein postulate $D_M\gamma^N = 0$ holds, G5). For the general warped metric FE-34, $\gamma^M\Omega_M = \frac{1}{2}\partial_\sigma(3\mathfrak{a} + 3\mathfrak{b} + \mathfrak{n})\gamma^{(0)} + \frac{1}{2}\mathrm{e}^{-\mathfrak{n}}\partial_{x_4}(3\mathfrak{a} + 3\mathfrak{b})\gamma^{(4)}$. Status: DERIVED.

4.2 The Higgs doublet

$$\begin{aligned}
&\frac{1}{\sqrt{-\hat h}\,\mathcal{Z}}\,D_\mu\big(\sqrt{-\hat h}\,\mathcal{Z}\,\hat h^{\mu\nu}D_\nu\Phi\big) - \mu^2\Phi - 2\lambda\big(\Phi^\dagger\Phi\big)\Phi - c_\Phi\,\bar\Psi P_\mathrm{L}\Psi\,\Phi = \Theta,\\
&\Theta = \begin{pmatrix} \bar d_\mathrm{R}(\mathcal{Y}^d)^\dagger u_\mathrm{L} + \bar e_\mathrm{R}(\mathcal{Y}^e)^\dagger \nu_\mathrm{L} - \bar d_\mathrm{L}\mathcal{Y}^u u_\mathrm{R} - \bar e_\mathrm{L}\mathcal{Y}^\nu \nu_\mathrm{R} \\ \bar d_\mathrm{R}(\mathcal{Y}^d)^\dagger d_\mathrm{L} + \bar e_\mathrm{R}(\mathcal{Y}^e)^\dagger e_\mathrm{L} + \bar u_\mathrm{L}\mathcal{Y}^u u_\mathrm{R} + \bar\nu_\mathrm{L}\mathcal{Y}^\nu \nu_\mathrm{R} \end{pmatrix}
\end{aligned} \tag{FE-15}$$

Symbols in (FE-15).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_{L}: The author's left-handed chiral projector, $P_{\mathrm{L}} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- Θ: The Yukawa source of the Higgs equation, $\Theta = -\partial\mathcal{L}_{\mathrm{Y}}/\partial\Phi^\dagger$ (a doublet of fermion bilinears). Unit: GeV^3. Status: STANDARD.
- d_{R}: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- u_{L}: The upper (up-type, left-handed) components of the quark doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- e_{R}: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_{L}: The upper (neutral, left-handed) components of the lepton doublets: the left-handed neutrinos. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_{L}: The lower (down-type, left-handed) components of the quark doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- u_{R}: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ν_{R}: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- e_{L}: The lower (charged, left-handed) components of the lepton doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

The field equation of the Higgs doublet on the fold: the variation of IN-11, IN-15 and IN-16 with respect to $\Phi^\dagger$. The factor $\mathcal{Z}$ (IN-8) is 1 in the late universe. The Higgs portal adds the term $-c_\Phi\bar{\Psi}P_{\mathrm{L}}\Psi\,\Phi$: the scalar density of the left-handed half of dirac16complex at the fold shifts the mass parameter of the Higgs field. Θ is the Yukawa source, $-\partial\mathcal{L}_{\mathrm{Y}}/\partial\Phi^\dagger$, written with the components $Q = (u_{\mathrm{L}}, d_{\mathrm{L}})$ and $\ell = (\nu_{\mathrm{L}}, e_{\mathrm{L}})$ of the doublets (generation indices contracted with the Yukawa matrices). Status: STANDARD (AUTHOR potential; INVENTED portal term).

4.3 The gauge fields

$$\frac{1}{\sqrt{-\hat{h}}\,\mathcal{Z}}\Big(\partial_\mu\big[\sqrt{-\hat{h}}\,\mathcal{Z}\,\mathcal{K}_G\,G^{A\mu\nu}\big] + g_3\,f^{ABC}G^B_\mu\sqrt{-\hat{h}}\,\mathcal{Z}\,\mathcal{K}_G\,G^{C\mu\nu}\Big) = -g_3\,J^{A\nu}_{\mathrm{c}}, \qquad \mathcal{K}_G = 1 + \sum_X c^G_X\,\bar{\Psi}P_X\Psi,$$
$$J^{A\nu}_{\mathrm{c}} = \bar{Q}\,\hat{\gamma}^\nu t^A Q + \bar{u}_{\mathrm{R}}\,\hat{\gamma}^\nu t^A u_{\mathrm{R}} + \bar{d}_{\mathrm{R}}\,\hat{\gamma}^\nu t^A d_{\mathrm{R}} \tag{FE-16}$$

Symbols in (FE-16).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its

first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.

- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathcal{K}_G$, $\mathcal{K}_W$, $\mathcal{K}_B$: The portal factors of the gauge kinetic terms, $\mathcal{K}_G = 1 + \sum_X c_X^G \bar{\Psi} P_X \Psi$ (likewise for W and B): the gauge couplings at the fold are divided by $\sqrt{\mathcal{K}}$. Unit: dimensionless. Value: 1 where dirac16complex vanishes. Status: DERIVED.
- G_μ^A, $G_{\mu\nu}^A$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling (the strong coupling g_s). Unit: dimensionless. Value: $g_3(m_Z) = 1.2177$ ($\alpha_s = 0.1180$). Status: STANDARD.
- f^{ABC}: The structure constants of $SU(3)$, $[t^A, t^B] = \mathrm{i} f^{ABC} t^C$. Unit: dimensionless. Status: STANDARD.
- B: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- C: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- $J_{\mathrm{c}}^{A\nu}$: The colour current of the quarks. Unit: GeV^3. Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_{\mathrm{L}}$ for $X = \mathrm{L}$, P_{R} for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_{\mathrm{L}}, d_{\mathrm{L}})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu \hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- t^A: The generators of $SU(3)$ on colour triplets, $t^A = \lambda_{\mathrm{GM}}^A/2$ (the Gell-Mann matrices halved), $\mathrm{tr}(t^A t^B) = \delta^{AB}/2$. Unit: dimensionless. Status: STANDARD.
- u_{R}: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_{R}: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

The Yang-Mills equation of the gluons (with the convention of the signature $(+,+,+,-)$: $\nabla_\mu F^{\mu\nu} = -J^\nu$ for a Lagrangian $-\frac{1}{4}F^2 + A_\mu J^\mu$). The gluon-kinetic portals multiply the gluon kinetic term by $\mathcal{K}_G$: the strong coupling at the fold is $g_3/\sqrt{\mathcal{K}_G}$ where dirac16complex is dense. The colour current $J_{\mathrm{c}}^{A\nu}$ of the quarks is the source. Status: STANDARD (INVENTED portal factor).

$$\frac{1}{\sqrt{-\hat{h}}\,\mathcal{Z}}\Big(\partial_\mu\big[\sqrt{-\hat{h}}\,\mathcal{Z}\,\mathcal{K}_W\,W^{I\mu\nu}\big] + g_2\,\epsilon^{IJK}W_\mu^J\sqrt{-\hat{h}}\,\mathcal{Z}\,\mathcal{K}_W\,W^{K\mu\nu}\Big) = -g_2\big(J_{\mathrm{w}}^{I\nu} + J_\Phi^{I\nu}\big),$$
$$\mathcal{K}_W = 1 + \sum_X c_X^W\,\bar{\Psi}P_X\Psi, \qquad J_{\mathrm{w}}^{I\nu} = \bar{Q}\,\hat{\gamma}^\nu\frac{\tau^I}{2}Q + \bar{\ell}\,\hat{\gamma}^\nu\frac{\tau^I}{2}\ell, \qquad J_\Phi^{I\nu} = \mathrm{i}\Big[(D^\nu\Phi)^\dagger\frac{\tau^I}{2}\Phi - \Phi^\dagger\frac{\tau^I}{2}D^\nu\Phi\Big] \tag{FE-17}$$

Symbols in (FE-17).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}_{\mu\nu}^{(1)}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.

- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathcal{K}_G$, $\mathcal{K}_W$, $\mathcal{K}_B$: The portal factors of the gauge kinetic terms, $\mathcal{K}_G = 1 + \sum_X c_X^G \bar\Psi P_X \Psi$ (likewise for W and B): the gauge couplings at the fold are divided by $\sqrt{\mathcal{K}}$. Unit: dimensionless. Value: 1 where dirac16complex vanishes. Status: DERIVED.
- W_μ^I, $W_{\mu\nu}^I$: The gauge field of $SU(2)_L$ on the fold, $W_\mu^1, W_\mu^2, W_\mu^3$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- ϵ^{IJK}: The Levi-Civita symbol of three indices (the structure constants of $SU(2)$), $\epsilon^{123} = 1$. Unit: dimensionless. Status: STANDARD.
- J: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- K: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- $J_{\mathrm{W}}^{I\nu}$: The weak-isospin current of the left-handed fermion doublets. Unit: GeV^3. Status: STANDARD.
- $J_\Phi^{I\nu}$: The weak-isospin current of the Higgs doublet (it contains the mass terms of the W fields); $J_\Phi^{2\,x_4}$ its isospin charge density with $I = 2$. Unit: GeV^3. Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- Q, $\bar Q$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge $1/6$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat\gamma^\mu = \hat{\mathbf{e}}_a{}^\mu \hat\gamma^{(a)}$, $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- ℓ, $\bar\ell$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat h^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge $1/2$), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.

The Yang-Mills equation of the $SU(2)_L$ fields. The sources are the weak-isospin current of the left-handed doublets and the isospin current of the Higgs doublet (which contains the mass terms of the W fields: in the vacuum $J_\Phi^{I\nu} \ni -g_2(v^2 + \mathcal{A}^2)W^{I\nu}/4$, so that FE-17 becomes $\nabla_\mu W^{I\mu\nu} = m_\mathrm{W}^2 W^{I\nu} + \ldots$). Status: STANDARD (INVENTED portal factor).

$$\begin{aligned} &\frac{1}{\sqrt{-\hat h}\,\mathcal{Z}}\,\partial_\mu\Big[\sqrt{-\hat h}\,\mathcal{Z}\Big(\mathcal{K}_B\,B^{\mu\nu} + \sum_X d_X\,\mathcal{D}_X^{\mu\nu}\Big)\Big] = -g_1\,J_Y^\nu, \qquad \mathcal{K}_B = 1 + \sum_X c_X^B\,\bar\Psi P_X \Psi, \\ &J_Y^\nu = \sum_\psi Y_\psi\,\bar\psi\,\hat\gamma^\nu\psi + \frac{\mathrm{i}}{2}\big[(D^\nu\Phi)^\dagger\Phi - \Phi^\dagger D^\nu\Phi\big] \end{aligned} \tag{FE-18}$$

Symbols in (FE-18).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathcal{K}_G$, $\mathcal{K}_W$, $\mathcal{K}_B$: The portal factors of the gauge kinetic terms, $\mathcal{K}_G = 1 + \sum_X c_X^G \bar{\Psi} P_X \Psi$ (likewise for W and B): the gauge couplings at the fold are divided by $\sqrt{\mathcal{K}}$. Unit: dimensionless. Value: 1 where dirac16complex vanishes. Status: DERIVED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar{\Psi}\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- g_1: The $U(1)_Y$ gauge coupling (usually g'; not normalised for grand unification). Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- J_Y^ν: The hypercharge current of the fermions and of the Higgs doublet. Unit: GeV^3. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$. Status: STANDARD.
- Y_f: The hypercharge of a field (IN-13: 1/6, 2/3, $-1/3$, $-1/2$, -1, 0 for $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$, 1/2 for Φ). Unit: dimensionless. Status: STANDARD.
- ψ, $\bar{\psi}$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar{\psi} = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu \hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.

The Maxwell equation of the hypercharge field. The dipole portal adds the dipole density of dirac16complex to the field strength, as a polarisation adds to the electric field in a medium: dirac16complex at the fold carries electric and magnetic dipole moments. The source is the hypercharge current of the fermions and of the Higgs doublet. Status: STANDARD (INVENTED portal terms).

4.4 The quarks and leptons

$$\mathrm{i}\hat{\gamma}^\mu\big(D_\mu + \mathrm{i}\,\mathcal{V}_\mu^Q\big)Q = \mathcal{Y}^u\tilde{\Phi}\,u_\mathrm{R} + \mathcal{Y}^d\Phi\,d_\mathrm{R}, \qquad \mathcal{V}_\mu^f = \sum_X c_X^f\, j_{X\mu} \tag{FE-19}$$

Symbols in (FE-19).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu \hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- u_R: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- d_R: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c^f_X: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind). Unit: GeV^{-7}. Status: INVENTED.
- j^μ_X: The convective current of the half X of dirac16complex, $j^\mu_X = \mathrm{i}(\bar{\Psi}P_X D^\mu\Psi - (D^\mu\bar{\Psi})P_X\Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.

The Dirac equation of the quark doublets: the variation of IN-14, IN-15 and IN-16 with respect to $\bar{Q}$. The current portals act on every Standard-Model fermion of kind f as a vector potential $\mathcal{V}^f_\mu$ built from the convective currents of the two halves of dirac16complex at the fold; it is the same for the three generations (flavour blind). Status: STANDARD (INVENTED potential).

$$\mathrm{i}\hat{\gamma}^\mu\big(D_\mu + \mathrm{i}\,\mathcal{V}^u_\mu\big)u_\mathrm{R} = (\mathcal{Y}^u)^\dagger\tilde{\Phi}^\dagger Q \tag{FE-20}$$

Symbols in (FE-20).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu \hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- u_R: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.

- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge $1/6$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

$$\mathrm{i}\hat{\gamma}^\mu\big(D_\mu + \mathrm{i}\,\mathcal{V}_\mu^d\big)d_\mathrm{R} = (\mathcal{Y}^d)^\dagger\Phi^\dagger Q \tag{FE-21}$$

Symbols in (FE-21).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}_\mu^f$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}_\mu^f = \sum_X c_X^f j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- d_R: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge $1/2$), a field of the fold. Unit: GeV. Status: AUTHOR.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge $1/6$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

$$\mathrm{i}\hat{\gamma}^\mu\big(D_\mu + \mathrm{i}\,\mathcal{V}_\mu^\ell\big)\ell = \mathcal{Y}^e\Phi\, e_\mathrm{R} + \mathcal{Y}^\nu\tilde{\Phi}\,\nu_\mathrm{R} \tag{FE-22}$$

Symbols in (FE-22).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ of IN-13; $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}_\mu^f$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}_\mu^f = \sum_X c_X^f j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge $1/2$), a field of the fold. Unit: GeV. Status: AUTHOR.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.

$$\mathrm{i}\hat{\gamma}^{\mu}\big(D_{\mu}+\mathrm{i}\,\mathcal{V}^{e}_{\mu}\big)e_{\mathrm{R}}=(\mathcal{Y}^{e})^{\dagger}\Phi^{\dagger}\ell \tag{FE-23}$$

Symbols in (FE-23).

- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^{\mu}$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^{\mu}=\hat{\mathbf{e}}_a{}^{\mu}\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^{\mu},\hat{\gamma}^{\nu}\}=-2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_{\mu}=\hat{h}_{\mu\nu}\hat{\gamma}^{\nu}$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1,x_2,x_3,x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_{μ}: The covariant derivative: on dirac16complex $D_M=\partial_M+\Omega_M$ (with $D_M\bar{\Psi}=\partial_M\bar{\Psi}-\bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_{μ} of IN-13; $D^{\mu}=\hat{h}^{\mu\nu}D_{\nu}$; on an adjoint field $(D_{\mu}X)^A=\partial_{\mu}X^A+g_3f^{ABC}G^B_{\mu}X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_{\mu}$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_{\mu}=\sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q,\mathcal{V}^u,\ldots,\mathcal{V}^{\nu}$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- e_{R}: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^{\nu}$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell=(\nu_{\mathrm{L}},e_{\mathrm{L}})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

$$\mathrm{i}\hat{\gamma}^{\mu}\big(D_{\mu}+\mathrm{i}\,\mathcal{V}^{\nu}_{\mu}\big)\nu_{\mathrm{R}}=(\mathcal{Y}^{\nu})^{\dagger}\tilde{\Phi}^{\dagger}\ell \tag{FE-24}$$

Symbols in (FE-24).

- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^{\mu}$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^{\mu}=\hat{\mathbf{e}}_a{}^{\mu}\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^{\mu},\hat{\gamma}^{\nu}\}=-2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_{\mu}=\hat{h}_{\mu\nu}\hat{\gamma}^{\nu}$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1,x_2,x_3,x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_{μ}: The covariant derivative: on dirac16complex $D_M=\partial_M+\Omega_M$ (with $D_M\bar{\Psi}=\partial_M\bar{\Psi}-\bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_{μ} of IN-13; $D^{\mu}=\hat{h}^{\mu\nu}D_{\nu}$; on an adjoint field $(D_{\mu}X)^A=\partial_{\mu}X^A+g_3f^{ABC}G^B_{\mu}X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_{\mu}$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_{\mu}=\sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q,\mathcal{V}^u,\ldots,\mathcal{V}^{\nu}$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- ν_{R}: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^{\nu}$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi}=\mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell=(\nu_{\mathrm{L}},e_{\mathrm{L}})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

The Dirac equations of the up-quark singlets, the down-quark singlets, the lepton doublets, the charged-lepton singlets and the right-handed neutrinos (FE-20 to FE-24), each the variation of IN-14, IN-15 and IN-16 with respect to the conjugate field. Each fermion sees the gauge fields (in D_{μ}), the Higgs doublet (through the Yukawa matrices) and dirac16complex (through $\mathcal{V}^f_{\mu}$). The right-handed neutrinos see no gauge field; they couple to the Standard Model only through the neutrino Yukawa matrix, which gives the neutrinos their Dirac masses. Status: STANDARD (INVENTED potentials).

$$\hat{\nabla}_\mu J^\mu_\ell = -\hat{\nabla}_\mu \big(J^\mu_e + J^\mu_\nu\big) = -\mathrm{i}\big(\bar{\ell}\,\mathcal{Y}^e \Phi\, e_\mathrm{R} + \bar{\ell}\,\mathcal{Y}^\nu \tilde{\Phi}\, \nu_\mathrm{R}\big) + \mathrm{h.c.},$$
$$\hat{\nabla}_\mu J^\mu_Q = -\hat{\nabla}_\mu \big(J^\mu_u + J^\mu_d\big) = -\mathrm{i}\big(\bar{Q}\,\mathcal{Y}^u \tilde{\Phi}\, u_\mathrm{R} + \bar{Q}\,\mathcal{Y}^d \Phi\, d_\mathrm{R}\big) + \mathrm{h.c.} \tag{FE-25}$$

Symbols in (FE-25).

- $\hat{\nabla}$: The covariant derivative of the fold: with the Levi-Civita connection of the metric of the fold ($\hat{h}_{\mu\nu}$ for four-dimensional tensors, the induced metric of the seven-dimensional fold for $\hat{\nabla}_a \mathcal{S}^a{}_b$). Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- J^μ_f: The fermion-number current of the Standard-Model multiplets of kind f, $J^\mu_f = \bar{\psi}_f \hat{\gamma}^\mu \psi_f$ summed over the three generations and the gauge components; with other labels the gauge currents (FE-16 to FE-18). Unit: GeV^3. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge $1/2$), a field of the fold. Unit: GeV. Status: AUTHOR.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2 \Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- h.c.: The Hermitian conjugate of the preceding terms. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge $1/6$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- u_R: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge $2/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_R: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

The divergences of the fermion-number currents of the six kinds of multiplets (DERIVED from FE-19 to FE-24): the Yukawa couplings turn doublets into singlets, so the current of each kind is not conserved by itself, while the lepton number $J_\ell + J_e + J_\nu$ and the baryon number $(J_Q + J_u + J_d)/3$ are conserved. The divergences enter the fold operator $\mathbb{M}$ (FE-13). Status: DERIVED.

4.5 The ad hoc Higgs vacuum

$$\mathcal{R}_\mathcal{A} = \mathcal{A}'' + 3\frac{b_4'}{b_4}\mathcal{A}' + \lambda\mathcal{A}^3 + c_\Phi\, \bar{\Psi} P_\mathrm{L} \Psi\, \mathcal{A} = \Big(4a_{40}^2 - 6a_{40}\frac{b_4'}{b_4}\Big)\mathcal{A} + \lambda\mathcal{A}^3 + c_\Phi\, \bar{\Psi} P_\mathrm{L} \Psi\, \mathcal{A},$$
$$\mathcal{R}_v = \lambda\mathcal{A}^2 v + c_\Phi\, \bar{\Psi} P_\mathrm{L} \Psi\, v, \qquad J_\Phi^{2\,x_4} = a_{40}\, v\, \mathcal{A}, \qquad J_\Phi^{1\,x_4} = J_\Phi^{3\,x_4} = 0 \tag{FE-26}$$

Symbols in (FE-26).

- $\mathcal{R}_\mathcal{A}$, $\mathcal{R}_v$: The residuals of the ad hoc Higgs vacuum in the upper ($\mathcal{R}_\mathcal{A}$) and lower ($\mathcal{R}_v$) components of the Higgs equation: what the ad hoc vacuum leaves unbalanced. Unit: GeV^3. Status: DERIVED.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_\mathcal{A} \mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.

- P_{L}: The author's left-handed chiral projector, $P_{\mathrm{L}} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- a_{40}: The decay constant of almightyAdHoc, $a_{40} > 0$: $\mathcal{A}$ falls by e in the time $1/(2a_{40})$. Unit: GeV (1/length). Status: AUTHOR.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV. Status: STANDARD.
- $J_\Phi^{I\nu}$: The weak-isospin current of the Higgs doublet (it contains the mass terms of the W fields); $J_\Phi^{2\,x_4}$ its isospin charge density with $I = 2$. Unit: GeV^3. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.

The author's ad hoc vacuum IN-12 inserted into the Higgs equation FE-15 (homogeneous fields, no gauge fields, the fold factor $\mathcal{Z} \propto b_4^3 \mathrm{e}^{-3a_4}$ so that the time derivative of $\sqrt{-\hat{h}}\mathcal{Z}$ gives $3b_4'/b_4$). The upper and lower components of FE-15 are not satisfied: $\mathcal{R}_\mathcal{A}$ and $\mathcal{R}_v$ are what remains (the residuals; FE-15 holds exactly if the source Θ is replaced by $\Theta - (\mathcal{R}_\mathcal{A}, \mathcal{R}_v)/\sqrt{2}$). The ad hoc vacuum is an exact solution only in the limit $\mathcal{A} \to 0$. It also carries a weak-isospin charge density $J_\Phi^{2\,x_4} = a_{40} v \mathcal{A}$ (the component along x_4 of the isospin current of FE-17 with $I = 2$), which the W^2 equation FE-17 must balance with a background field $W_{x_4}^2$ (DERIVED, `um3_fields.py`, F5: the masses of the W and Z, the direction of the photon, the charge density, the residual of the upper component). Status: DERIVED.

5 The background

The background is the author's metric with the two functions $a_4(x_4)$ and $b_4(x_4)$ of the time, on which every other field moves; the matter fields correct it by the next terms of a power series (FE-35). This section writes out its geometry and the exact equations that its two functions obey, in the bulk and at the fold.

5.1 The author's metric and the proper chart

$$
\begin{aligned}
g_{MN} &= \mathrm{diag}\Big(\cot^2 z,\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ -1, \\
&\qquad -\mathrm{e}^{-2a_4} b_4^2 (\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4} b_4^2 (\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4} b_4^2 (\sin^2 z)^{1/6} \Big), \qquad z = 6Hx_0, \\
\sigma &= \mathrm{sgn}\Big(\frac{\pi}{2} - z\Big) \frac{\ln \sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\, \mathrm{d}x_0, \\
g_{MN}\, \mathrm{d}X^M \mathrm{d}X^N &= \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|} \Big[\mathrm{e}^{2a_4} \big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2 \big) - \mathrm{e}^{-2a_4} b_4^2 \big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2 \big) \Big] - \mathrm{d}x_4^2
\end{aligned}
\tag{FE-27}
$$

Symbols in (FE-27).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\, \mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\, \mathrm{d}^8 X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The author's metric of the primordial gravitational field (his order of 2026-10-24: the metric of `Dirac_Einstein-Lovelock_4+4_Nash.nb` with $b_4 \to b_4^2$; $(\sin^2 z)^{1/6}$ is the positive branch of `Sin[6 H x0]^(1/3)`, the author's rule of 2026-10-16), and the same metric in the proper chart σ of the hidden direction. In the author's chart the metric is smooth but degenerate at the fold $z = \pi/2$ ($\cot z = 0$); the proper distance from the fold is $|\sigma|$, and σ runs from $-\infty$ (the tip $z \to 0$) through 0 (the fold) to $+\infty$ (the tip $z \to \pi$): one period of x_0 is the whole line, and the two halves of a period are mirror images under $z \to \pi - z$ ($\sigma \to -\sigma$). Ordinary space and the extra times are warped by $\mathrm{e}^{-H|\sigma|}$ (with a kink at the fold), the time x_4 is not warped. a_4 and b_4 are functions of x_4; b_4 is never identically 1. Status: AUTHOR (the metric), DERIVED (the chart, `um3_fold.py`, J1).

$$
\begin{aligned}
&\mathbf{e}^{(0)} = \mathrm{d}\sigma, \quad \mathbf{e}^{(i)} = \mathrm{e}^{a_4 - H|\sigma|}\mathrm{d}x_i, \quad \mathbf{e}^{(4)} = \mathrm{d}x_4, \quad \mathbf{e}^{(r)} = b_4\,\mathrm{e}^{-a_4 - H|\sigma|}\mathrm{d}x_r,\\
&\sqrt{|g|} = b_4^3\,\mathrm{e}^{-6H|\sigma|}, \qquad \mathsf{A} = \frac{a_4'}{H}, \qquad \mathsf{U} = \frac{b_4'}{Hb_4}, \qquad \mathsf{A}_1 = \frac{\mathsf{A}'}{H}, \qquad \mathsf{U}_1 = \frac{\mathsf{U}'}{H},\\
&R^{(0)(i)}{}_{(0)(i)} = R^{(0)(r)}{}_{(0)(r)} = -H^2, \qquad R^{(0)(4)}{}_{(0)(4)} = 0, \qquad R^{(i)(j)}{}_{(i)(j)} = (\mathsf{A}^2 - 1)H^2,\\
&R^{(i)(4)}{}_{(i)(4)} = (\mathsf{A}^2 + \mathsf{A}_1)H^2, \qquad R^{(i)(r)}{}_{(i)(r)} = -(1 + \mathsf{A}^2 - \mathsf{AU})H^2,\\
&R^{(4)(r)}{}_{(4)(r)} = \big((\mathsf{U} - \mathsf{A})^2 + \mathsf{U}_1 - \mathsf{A}_1\big)H^2, \qquad R^{(r)(s)}{}_{(r)(s)} = \big((\mathsf{U} - \mathsf{A})^2 - 1\big)H^2,\\
&R^{(0)(i)}{}_{(4)(i)} = \mathrm{sgn}(\sigma)\,\mathsf{A}H^2, \qquad R^{(0)(r)}{}_{(4)(r)} = \mathrm{sgn}(\sigma)\,(\mathsf{U} - \mathsf{A})H^2
\end{aligned}
\tag{FE-28}
$$

Symbols in (FE-28).

- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

- x: The coordinates of the fold: x_i ($i = 1, 2, 3$) of ordinary space, x_4 the time, x_r ($r = 5, 6, 7$) the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- ${R^{MN}}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor (convention $[\nabla_P, \nabla_Q]V^M = {R^M}_{NPQ}V^N$), with two indices the Ricci tensor $R_{MN} = {R^P}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices ${R^{(a)(b)}}_{(c)(d)}$ its frame components, ${R^{(a)(b)}}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- j: A second generation or mass-state index (as i). Status: STANDARD.
- s: A second extra-time direction of the fold (as r, different from it). Status: STANDARD.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.

The vielbein and the volume element (i, j any two different directions of ordinary space, r, s any two different extra times; x_i and x_r the coordinates of these directions), the four dimensionless rates of the background (A the expansion rate of ordinary space in units of H, U the rate of change of b_4), and every non-zero frame component of the Riemann tensor, up to the symmetries ${R^{ab}}_{cd} = -{R^{ba}}_{cd} = -{R^{ab}}_{dc} = {R_{cd}}^{ab}$ (the last with the frame metric). No component depends on σ except through the sign of the two mixed ones, which change sign at the fold: the bulk on each side of the fold is homogeneous along σ (DERIVED, `um3_geometry.py`, G2, exact; `um3_fold.py`, J2). The frame curvature of the hidden direction is $-H^2$: each half of a period is a space of constant negative curvature along σ, like a slice of anti-de Sitter space, with the time not warped. Status: DERIVED.

5.2 The bulk equations of the background

$$\frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_{(0)}}{H^2} = -\frac{1}{16}\Big(\mathsf{A}^6 - 3\mathsf{A}^5\mathsf{U} + 12\mathsf{A}^4\mathsf{U}^2 + 3\mathsf{A}^4\mathsf{U}_1 + 3\mathsf{A}^4 + 12\mathsf{A}^3\mathsf{A}_1\mathsf{U} - 19\mathsf{A}^3\mathsf{U}^3 - 12\mathsf{A}^3\mathsf{U}\mathsf{U}_1 - 6\mathsf{A}^3\mathsf{U} - 18\mathsf{A}^2\mathsf{A}_1\mathsf{U}^2 + 9\mathsf{A}^2\mathsf{U}^4 + 9\mathsf{A}^2\mathsf{U}^2\mathsf{U}_1 - 18\mathsf{A}^2\mathsf{U}^2 - 6\mathsf{A}^2\mathsf{U}_1 - 9\mathsf{A}^2 + 6\mathsf{A}\mathsf{A}_1\mathsf{U}^3 - 12\mathsf{A}\mathsf{A}_1\mathsf{U} + 21\mathsf{A}\mathsf{U}^3 + 12\mathsf{A}\mathsf{U}\mathsf{U}_1 + 9\mathsf{A}\mathsf{U} + 6\mathsf{A}_1\mathsf{U}^2 + 3\mathsf{U}^4 + 3\mathsf{U}^2\mathsf{U}_1 + 18\mathsf{U}^2 + 3\mathsf{U}_1 + 5\Big),$$

$$\frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(i)}{}_{(i)}}{H^2} = -\frac{1}{16}\Big(\mathsf{A}^6 - 6\mathsf{A}^5\mathsf{U} - 5\mathsf{A}^4\mathsf{A}_1 + 12\mathsf{A}^4\mathsf{U}^2 + 3\mathsf{A}^4\mathsf{U}_1 + 3\mathsf{A}^4 + 12\mathsf{A}^3\mathsf{A}_1\mathsf{U} - 10\mathsf{A}^3\mathsf{U}^3 - 6\mathsf{A}^3\mathsf{U}\mathsf{U}_1 + 12\mathsf{A}^3\mathsf{U} - 9\mathsf{A}^2\mathsf{A}_1\mathsf{U}^2 + 6\mathsf{A}^2\mathsf{A}_1 + 3\mathsf{A}^2\mathsf{U}^4 + 3\mathsf{A}^2\mathsf{U}^2\mathsf{U}_1 - 36\mathsf{A}^2\mathsf{U}^2 - 6\mathsf{A}^2\mathsf{U}_1 - 9\mathsf{A}^2 + 2\mathsf{A}\mathsf{A}_1\mathsf{U}^3 - 12\mathsf{A}\mathsf{A}_1\mathsf{U} + 18\mathsf{A}\mathsf{U}^3 + 6\mathsf{A}\mathsf{U}\mathsf{U}_1 - 6\mathsf{A}\mathsf{U} + 3\mathsf{A}_1\mathsf{U}^2 - \mathsf{A}_1 + 3\mathsf{U}^4 + 3\mathsf{U}^2\mathsf{U}_1 + 24\mathsf{U}^2 + 3\mathsf{U}_1 + 5\Big),$$

$$\frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(4)}{}_{(4)}}{H^2} = \frac{1}{16}\Big(5\mathsf{A}^6 - 15\mathsf{A}^5\mathsf{U} + 15\mathsf{A}^4\mathsf{U}^2 - 9\mathsf{A}^4 - 5\mathsf{A}^3\mathsf{U}^3 + 18\mathsf{A}^3\mathsf{U} + 3\mathsf{A}^2 - 9\mathsf{A}\mathsf{U}^3 - 3\mathsf{A}\mathsf{U} - 3\mathsf{U}^2 + 1\Big),$$

$$\frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(r)}{}_{(r)}}{H^2} = -\frac{1}{16}\Big(\mathsf{A}^6 + 5\mathsf{A}^4\mathsf{A}_1 - 3\mathsf{A}^4\mathsf{U}^2 - 2\mathsf{A}^4\mathsf{U}_1 + 3\mathsf{A}^4 - 8\mathsf{A}^3\mathsf{A}_1\mathsf{U} + 2\mathsf{A}^3\mathsf{U}^3 + 2\mathsf{A}^3\mathsf{U}\mathsf{U}_1 - 24\mathsf{A}^3\mathsf{U} + 3\mathsf{A}^2\mathsf{A}_1\mathsf{U}^2 - 6\mathsf{A}^2\mathsf{A}_1 + 18\mathsf{A}^2\mathsf{U}^2 - 9\mathsf{A}^2 + 6\mathsf{A}\mathsf{U}^3 + 6\mathsf{A}\mathsf{U}\mathsf{U}_1 + 24\mathsf{A}\mathsf{U} + 3\mathsf{A}_1\mathsf{U}^2 + \mathsf{A}_1 + 9\mathsf{U}^2 + 2\mathsf{U}_1 + 5\Big),$$

$$\frac{\kappa\,{\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_{(4)}}{H^2} = \mathrm{sgn}(\sigma)\,\frac{3}{16}\,\mathsf{U}\,\mathcal{G}, \qquad \mathcal{G} = 5\mathsf{A}^2(\mathsf{A}-\mathsf{U})^2 - 6\mathsf{A}^2 + 6\mathsf{A}\mathsf{U} + \mathsf{U}^2 + 1 \tag{FE-29}$$

Symbols in (FE-29).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{stab}}$; ${\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\mathcal{G}$: The factor of the (σ, x_4) bulk equation of the author's metric, $\mathcal{G} = 5\mathsf{A}^2(\mathsf{A}-\mathsf{U})^2 - 6\mathsf{A}^2 + 6\mathsf{A}\mathsf{U} + \mathsf{U}^2 + 1$. Unit: dimensionless. Status: DERIVED.

The bulk equations FE-1 for the author's metric, divided by H^2: the left-hand side of FE-1 with the author's weights and $\Lambda = -10H^2$, evaluated exactly (DERIVED, `um3_geometry.py`, G3 and G4, with the identities FE-3 checked; `um3_fold.py`, J2, by an independent general computation). They state what the bulk sources must be for given $a_4(x_4)$, $b_4(x_4)$: ${\mathcal{T}^{(4)}}_{(4)} = -\rho$ gives the energy density, ${\mathcal{T}^{(0)}}_{(0)}$, ${\mathcal{T}^{(i)}}_{(i)}$, ${\mathcal{T}^{(r)}}_{(r)}$ the pressures along σ, ordinary space and the extra times, and ${\mathcal{T}^{(0)}}_{(4)}$ the energy flux along σ, which has opposite signs on the two sides of the fold (it flows toward or away from the fold on both sides). Because the frame components do not depend on σ, the author's metric can be exact only with sources whose frame components do not depend on σ; sources that depend on σ require the next terms of the power series of the metric (FE-35). The time-time equation contains no A_1 and no U_1: it is the energy constraint of the background. The flux equation factorises: with a source without flux along σ, either $\mathsf{U} = 0$ (b_4 constant) or $\mathcal{G}(\mathsf{A}, \mathsf{U}) = 0$, which has real roots U only for $0.3249 \le |\mathsf{A}| \le 1.3764$ ($5\mathsf{A}^4 - 10\mathsf{A}^2 + 1 \le 0$). Status: DERIVED.

5.3 The fold equations of the background

$$\begin{aligned}
\frac{\kappa\,\mathcal{S}^{(i)}{}_{(i)}}{H} &= \frac{1}{24}\Big(9\mathsf{A}^4 - 24\mathsf{A}^3\mathsf{U} - 12\mathsf{A}^2\mathsf{A}_1 + 27\mathsf{A}^2\mathsf{U}^2 + 12\mathsf{A}^2\mathsf{U}_1 - 66\mathsf{A}^2 + 24\mathsf{A}\mathsf{A}_1\mathsf{U} - 6\mathsf{A}\mathsf{U}^3 - 12\mathsf{A}\mathsf{U}\mathsf{U}_1 + 120\mathsf{A}\mathsf{U} \\
&\qquad - 6\mathsf{A}_1\mathsf{U}^2 + 20\mathsf{A}_1 - 6\mathsf{U}^4 - 6\mathsf{U}^2\mathsf{U}_1 - 111\mathsf{U}^2 - 60\mathsf{U}_1 - 183\Big), \\
\frac{\kappa\,\mathcal{S}^{(4)}{}_{(4)}}{H} &= -\frac{1}{4}\Big(3\mathsf{A}^4 - 6\mathsf{A}^3\mathsf{U} - 10\mathsf{A}^2 + 3\mathsf{A}\mathsf{U}^3 + 10\mathsf{A}\mathsf{U} + 10\mathsf{U}^2 + 31\Big), \\
\frac{\kappa\,\mathcal{S}^{(r)}{}_{(r)}}{H} &= \frac{1}{24}\Big(9\mathsf{A}^4 - 12\mathsf{A}^3\mathsf{U} + 12\mathsf{A}^2\mathsf{A}_1 + 9\mathsf{A}^2\mathsf{U}^2 - 66\mathsf{A}^2 - 12\mathsf{A}\mathsf{U}^3 - 12\mathsf{A}\mathsf{U}\mathsf{U}_1 + 12\mathsf{A}\mathsf{U} - 6\mathsf{A}_1\mathsf{U}^2 - 20\mathsf{A}_1 \\
&\qquad - 57\mathsf{U}^2 - 40\mathsf{U}_1 - 183\Big)
\end{aligned} \tag{FE-30}$$

Symbols in (FE-30).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.

The junction conditions FE-4 for the author's metric: the thin source that the fold must carry for given $a_4(x_4)$, $b_4(x_4)$ (the off-diagonal and (σ,σ) components have no thin part). They follow from the potential $\mathfrak{F}$ evaluated with $\partial_\sigma\mathfrak{a} = \partial_\sigma\mathfrak{b} = -H\,\mathrm{sgn}(\sigma)$ on the two sides (DERIVED, `um3_fold.py`, J4 and J5, and independently by smoothing the kink, $\partial_\sigma\mathfrak{a} = -H\tanh(\sigma/\varepsilon)$, and integrating the full left-hand side of FE-1 across the layer: the result converges linearly in ε to FE-30). With the bulk flux of FE-29 they satisfy the conservation law FE-11 identically (DERIVED, J6). Status: DERIVED.

5.4 The author's vacuum

$$\begin{aligned}
&a_4 = Hx_4, \quad b_4 = \text{constant } (\mathsf{A} = 1,\ \mathsf{U} = \mathsf{A}_1 = \mathsf{U}_1 = 0): \qquad \sum_{p=1}^{3} \alpha_{(p)}\,\mathcal{E}_{(p)}{}^{M}{}_{N} + \Lambda\,\delta^M_N = 0 \quad (\sigma \neq 0), \\
&\kappa\,\mathcal{S}^{(i)}{}_{(i)} = \kappa\,\mathcal{S}^{(r)}{}_{(r)} = -10H, \qquad \kappa\,\mathcal{S}^{(4)}{}_{(4)} = -6H, \\
&\kappa\rho_{\mathrm{bulk}} = -H^2\Big[\frac{3}{4}\big((\mathsf{A} - 1 - \mathsf{U})^2 + (\mathsf{A} - 1)^2\big) + \ldots\Big], \qquad \mathcal{G}(1, \mathsf{U}) = 2\mathsf{U}(3\mathsf{U} - 2)
\end{aligned} \tag{FE-31}$$

Symbols in (FE-31).

- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.

- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1 = \mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- ${\mathcal{E}_{(p)}}^M{}_N$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -10H^2$ (the author writes the pure number $\Lambda_\mathrm{N} = 10$ in $0 = -\Lambda_\mathrm{N} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-10H^2$. Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ${\mathcal{S}^a}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- ρ_bulk: The energy density of the bulk sources, $\rho_\mathrm{bulk} = -{\mathcal{T}_\mathrm{bulk}}^{(4)}{}_{(4)}$. Unit: GeV^8. Status: DERIVED.
- $\mathcal{G}$: The factor of the (σ, x_4) bulk equation of the author's metric, $\mathcal{G} = 5\mathsf{A}^2(\mathsf{A} - \mathsf{U})^2 - 6\mathsf{A}^2 + 6\mathsf{A}\mathsf{U} + \mathsf{U}^2 + 1$. Unit: dimensionless. Status: DERIVED.

The author's solution (`Dirac_Einstein-Lovelock_4+4_Nash.nb`, In[134]: $a_4' = H$, the weights $(1, 1/32, 1/1152)$ and $\Lambda_\mathrm{N} = 10$): ordinary space inflates with the rate H forever and the extra times deflate ($\mathrm{e}^{-a_4} b_4 \propto \mathrm{e}^{-Hx_4}$), "eternal inflation/deflation". With the new metric (b_4^2) it is still an exact vacuum solution of the bulk equations for every constant b_4 (DERIVED, `um3_geometry.py`, G4). Three facts about it (DERIVED):

- The fold must carry a thin source: $\kappa\mathcal{S} = \mathrm{diag}(-10H, -6H, -10H)$ for (ordinary space, time, extra times), an energy density $6H/\kappa$ with the pressure $-10H/\kappa$ (`um3_fold.py`, J5).
- The vacuum is degenerate: near it the energy density of any bulk source is $\kappa\rho_\mathrm{bulk} = -H^2[\frac{3}{4}((\mathsf{A} - 1 - \mathsf{U})^2 + (\mathsf{A} - 1)^2) + \ldots]$ (the dots are of third order in $\mathsf{A} - 1$ and U): the linear terms vanish, so a small positive energy density of the bulk cannot be carried by a small change of a_4 and b_4 (`um3_geometry.py`, G4). The transverse traceless gravitons of the bulk have no time-kinetic term at this point (`um3_bulk.py`, B5).
- b_4 is a function of x_4, never identically 1 (the author's order): a constant $b_4 \neq 1$ is equivalent to $b_4 = 1$ by a rescaling of x_5, x_6, x_7 unless their coordinate period is fixed, and $\mathsf{U} \neq 0$ is not a vacuum solution near $\mathsf{A} = 1$: it needs a bulk source with a flux along σ, $\kappa{\mathcal{T}^{(0)}}_{(4)} = \mathrm{sgn}(\sigma)\frac{3}{16}\mathsf{U}\,\mathcal{G}(1, \mathsf{U})H^2 = \mathrm{sgn}(\sigma)\frac{3}{8}\mathsf{U}^2(3\mathsf{U} - 2)H^2$, and a non-positive energy density. At $\mathsf{A} = 1$ the flux vanishes for $\mathsf{U} = 2/3$ ($b_4 \propto \mathrm{e}^{2Hx_4/3}$), which needs the constant source $\kappa\mathcal{T} = (-14/27, -2/9, 2/27, -22/27)H^2$ for (σ, ordinary space, time, extra times), with $\kappa\rho_\mathrm{bulk} = -2H^2/27$ (`um3_geometry.py`, G4).

Status: AUTHOR (the solution), DERIVED (the three facts).

5.5 The static block and the frozen extra times

$$\begin{aligned}&\mathsf{A}=\mathsf{U}=\mathsf{A}_1=\mathsf{U}_1=0: \qquad \kappa\,{\mathcal{T}_{\rm bulk}}^{a}{}_{b}=H^2\,\mathrm{diag}\Big(-\frac{5}{16},-\frac{5}{16},-\frac{5}{16},-\frac{5}{16},\frac{1}{16},-\frac{5}{16},-\frac{5}{16},-\frac{5}{16}\Big),\\&\kappa\,{\mathcal{S}^{(i)}}_{(i)}=\kappa\,{\mathcal{S}^{(r)}}_{(r)}=-\frac{61}{8}H, \qquad \kappa\,{\mathcal{S}^{(4)}}_{(4)}=-\frac{31}{4}H\end{aligned} \tag{FE-32}$$

Symbols in (FE-32).

- A: The expansion rate of ordinary space in units of H, $\mathsf{A}=a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U}=b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U}=\mathsf{A}$ with frozen extra times. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1=\mathsf{A}'/H=a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1=\mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa=8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\rm bulk}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi+\mathcal{T}_{\rm stab}$; ${\mathcal{T}_{\rm bulk}}^{(0)}{}_{b}$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- a: A frame (orthonormal) index running over $0,1,\ldots,7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a,b\in\{1,\ldots,7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda=-10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0,x_1,\ldots,x_7$ for a tensor of eight dimensions, $x_1,\ldots,x_7$ for a tensor of the fold, $x_1,\ldots,x_4$ for a four-dimensional tensor). Status: STANDARD.
- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- i: A generation index ($i=1,2,3$), which also labels the neutrino mass states ν_1,ν_2,ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r\in\{5,6,7\}$ (x_5,x_6,x_7); not summed (the three components are equal). Status: STANDARD.

The static block: a_4 and b_4 constant. Its bulk is (each half of a period) a seven-dimensional space of constant negative curvature $-H^2$ of the directions $\sigma,x_1,x_2,x_3,x_5,x_6,x_7$ times the unwarped time (FE-28 with $\mathsf{A}=\mathsf{U}=0$). It is not a vacuum solution: it requires the bulk source of the first line, which is the stabiliser IN-7 ($\kappa\rho=-H^2/16$, $\kappa p=-5H^2/16$), and the thin source of the second line at the fold, which is the fold fluid at rest ($\kappa\rho_{\rm ff}/V_*=31H/4$, $\kappa p_{\rm ff}/V_*=\kappa q_{\rm ff}/V_*=-61H/8$, $\rho+p=H/(8\kappa)$ per V_*, positive). The late universe, whose expansion rate is tiny in units of H, is the static block plus small corrections (FE-33, FE-59). Status: DERIVED.

$$\begin{aligned}&\mathsf{U}=\mathsf{A},\quad \mathsf{U}_1=\mathsf{A}_1\ \ \big(\mathrm{e}^{-a_4}b_4=\text{constant},\ \mathcal{Z}=1\big):\\&\frac{\kappa\rho_{\rm bulk}}{H^2}=\frac{3\mathsf{A}^2-1}{16}, \qquad \frac{\kappa\,{\mathcal{T}_{\rm bulk}}^{(0)}{}_{(0)}}{H^2}=-\frac{3\mathsf{A}^4+3\mathsf{A}^2\mathsf{A}_1+18\mathsf{A}^2+3\mathsf{A}_1+5}{16},\\&\frac{\kappa\,{\mathcal{T}_{\rm bulk}}^{(i)}{}_{(i)}}{H^2}=-\frac{9\mathsf{A}^2+2\mathsf{A}_1+5}{16}, \qquad \frac{\kappa\,{\mathcal{T}_{\rm bulk}}^{(r)}{}_{(r)}}{H^2}=-\frac{3\mathsf{A}^4+3\mathsf{A}^2\mathsf{A}_1+24\mathsf{A}^2+3\mathsf{A}_1+5}{16},\\&\frac{\kappa\,{\mathcal{T}_{\rm bulk}}^{(0)}{}_{(4)}}{H^2}=\mathrm{sgn}(\sigma)\,\frac{3\mathsf{A}(1+\mathsf{A}^2)}{16}, \qquad \frac{\kappa\,{\mathcal{S}^{(i)}}_{(i)}}{H}=-\frac{57\mathsf{A}^2+40\mathsf{A}_1+183}{24},\\&\frac{\kappa\,{\mathcal{S}^{(4)}}_{(4)}}{H}=-\frac{10\mathsf{A}^2+31}{4}, \qquad \frac{\kappa\,{\mathcal{S}^{(r)}}_{(r)}}{H}=-\frac{2\mathsf{A}^4+2\mathsf{A}^2\mathsf{A}_1+37\mathsf{A}^2+20\mathsf{A}_1+61}{8}\end{aligned} \tag{FE-33}$$

Symbols in (FE-33).

- U: The rate of change of b_4 in units of H, $\mathsf{U}=b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U}=\mathsf{A}$ with frozen extra times. Status: DERIVED.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A}=a_4'/H$. Unit: dimensionless. Value: 1 at the author's vacuum; H_0/H today (H_0 the Hubble constant). Status: DERIVED.
- U_1: The time derivative of U in units of H, $\mathsf{U}_1=\mathsf{U}'/H$. Unit: dimensionless. Status: DERIVED.

- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_{\mathrm{t}}(\mathrm{e}^{-a_4} b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ρ_{bulk}: The energy density of the bulk sources, $\rho_{\mathrm{bulk}} = -{\mathcal{T}_{\mathrm{bulk}}}^{(4)}{}_{(4)}$. Unit: GeV^8. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- $\mathcal{T}_{\mathrm{bulk}}$: The energy-momentum tensor of all sources of the bulk, $\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{stab}}$; ${\mathcal{T}_{\mathrm{bulk}}}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- ${\mathcal{S}^a}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.

The frozen extra times: the physical size $\mathrm{e}^{-a_4} b_4$ of the extra times is constant while ordinary space expands. Then the fold factor $\mathcal{Z}$ is constant (IN-8) and every coupling of the Standard Model is constant in time, as observation requires (the fine-structure constant has not changed by more than parts in 10^6 since the CMB); this is the condition of the late universe. It gives $b_4 = c_b\,\mathrm{e}^{a_4}$ with a constant c_b: b_4 grows with the scale factor of ordinary space, so it is a non-trivial function of x_4, never identically 1, as the author requires. The equations of FE-29 and FE-30 on this branch are written out (DERIVED, `um3_geometry.py`, G4, `um3_fold.py`, J7). For $\mathsf{A} \to 0$ they reduce to the static block FE-32. The extra energy of the fold above the static value is $(5/2)a_4'^2/(\kappa H)$ per V_*: on the separable metric the fold energy fixes the expansion rate; a separately conserved fold fluid fits the separable metric only for the equation of state $p = -\rho/40$ (DERIVED, J7), so ordinary matter on the fold requires the next terms of the power series (FE-35, FE-59). Status: DERIVED.

5.6 The general warped metric and the power series of the metric

$$
\begin{aligned}
&g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{d}\sigma^2 + \mathrm{e}^{2\mathfrak{a}}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{e}^{2\mathfrak{n}}\,\mathrm{d}x_4^2 - \mathrm{e}^{2\mathfrak{b}}\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big),\\
&R^{(0)(i)}{}_{(0)(i)} = -\partial_\sigma^2\mathfrak{a} - (\partial_\sigma\mathfrak{a})^2, \qquad R^{(0)(4)}{}_{(0)(4)} = -\partial_\sigma^2\mathfrak{n} - (\partial_\sigma\mathfrak{n})^2, \qquad R^{(i)(j)}{}_{(i)(j)} = -(\partial_\sigma\mathfrak{a})^2 + \mathrm{e}^{-2\mathfrak{n}}(\partial_{x_4}\mathfrak{a})^2,\\
&R^{(i)(r)}{}_{(i)(r)} = -\partial_\sigma\mathfrak{a}\,\partial_\sigma\mathfrak{b} + \mathrm{e}^{-2\mathfrak{n}}\,\partial_{x_4}\mathfrak{a}\,\partial_{x_4}\mathfrak{b}, \qquad R^{(i)(4)}{}_{(i)(4)} = -\partial_\sigma\mathfrak{a}\,\partial_\sigma\mathfrak{n} + \mathrm{e}^{-2\mathfrak{n}}\big(\partial_{x_4}^2\mathfrak{a} + (\partial_{x_4}\mathfrak{a})^2 - \partial_{x_4}\mathfrak{a}\,\partial_{x_4}\mathfrak{n}\big),\\
&R^{(0)(i)}{}_{(4)(i)} = -\mathrm{e}^{-\mathfrak{n}}\big(\partial_\sigma\partial_{x_4}\mathfrak{a} + \partial_\sigma\mathfrak{a}\,\partial_{x_4}\mathfrak{a} - \partial_{x_4}\mathfrak{a}\,\partial_\sigma\mathfrak{n}\big)
\end{aligned}
\tag{FE-34}
$$

Symbols in (FE-34).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\mathfrak{a}$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a} = a_4 - H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35. Unit: dimensionless. Status: DERIVED.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\mathfrak{n}$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4x_4} = -\mathrm{e}^{2\mathfrak{n}}$); for the author's metric $\mathfrak{n} = 0$; $\mathfrak{n}_n$ its term of order n. Unit: dimensionless. Status: DERIVED.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\mathfrak{b}$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$; $\mathfrak{b}_n$ its term of order n. Unit: dimensionless. Status: DERIVED.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor (convention $[\nabla_P, \nabla_Q]V^M = R^M{}_{NPQ}V^N$), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the neutrino mass states ν_1, ν_2, ν_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- j: A second generation or mass-state index (as i). Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.

The general metric with the symmetries of the author's metric (homogeneous and isotropic in ordinary space and in the extra times, independent of the extra times) but with three arbitrary functions $\mathfrak{a}$, $\mathfrak{b}$, $\mathfrak{n}$ of σ and x_4, and its frame curvature: the components with the extra times follow from those with ordinary space by replacing $\mathfrak{a}$ by $\mathfrak{b}$ and (i) by (r) ($R^{(0)(r)}{}_{(0)(r)}$, $R^{(r)(s)}{}_{(r)(s)}$, $R^{(r)(4)}{}_{(r)(4)}$, $R^{(0)(r)}{}_{(4)(r)}$); all others vanish (DERIVED, `um3_fold.py`, J2). Its Lovelock tensor is FE-2 with these components; it has only diagonal components and the (σ, x_4) pair. The author's metric is $\mathfrak{a} = a_4 - H|\sigma|$, $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$, $\mathfrak{n} = 0$, and the general equations reduce to FE-29 on both sides of the fold (DERIVED, J2). Status: DERIVED.

$$\begin{aligned} &\mathfrak{a} = a_4 - H|\sigma| + \sum_{n\geq 1} \epsilon_{\mathrm{r}}^{\,n}\, \mathfrak{a}_n(\sigma, x_4), \qquad \mathfrak{b} = \ln b_4 - a_4 - H|\sigma| + \sum_{n\geq 1} \epsilon_{\mathrm{r}}^{\,n}\, \mathfrak{b}_n(\sigma, x_4), \\ &\mathfrak{n} = \sum_{n\geq 1} \epsilon_{\mathrm{r}}^{\,n}\, \mathfrak{n}_n(\sigma, x_4), \qquad \hat{h}_{\mu\nu} \to \hat{h}_{\mu\nu} + \epsilon_{\mathrm{c}}\, \hat{h}^{(1)}_{\mu\nu} + \ldots \end{aligned} \tag{FE-35}$$

Symbols in (FE-35).

- $\mathfrak{a}$: The logarithm of the scale factor of ordinary space in the general warped metric FE-34, a function of σ and x_4; for the author's metric $\mathfrak{a} = a_4 - H|\sigma|$; $\mathfrak{a}_n$ its term of order n in the power series FE-35. Unit: dimensionless. Status: DERIVED.

- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- n: The order $n = 1, 2, \ldots$ of a term of the power series of the metric (also its exponent of ϵ_r); $\mathfrak{a}_n$, $\mathfrak{b}_n$, $\mathfrak{n}_n$ are the terms of order n. Status: STANDARD.
- ϵ_r: The bookkeeping parameter of the response of the bulk to the matter of the fold: the ratio of the bulk to the fold gravity, $\epsilon_\mathrm{r} \sim r = \kappa_4 V_*/(\kappa H)$. Unit: dimensionless. Value: $\lesssim 10^{-15}$. Status: DERIVED.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\mathfrak{b}$: The logarithm of the scale factor of the extra times in the general warped metric FE-34; for the author's metric $\mathfrak{b} = \ln b_4 - a_4 - H|\sigma|$; $\mathfrak{b}_n$ its term of order n. Unit: dimensionless. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4} b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- $\mathfrak{n}$: The logarithm of the lapse in the general warped metric FE-34 ($g_{x_4 x_4} = -\mathrm{e}^{2\mathfrak{n}}$); for the author's metric $\mathfrak{n} = 0$; $\mathfrak{n}_n$ its term of order n. Unit: dimensionless. Status: DERIVED.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- ϵ_c: The bookkeeping parameter of the cosmological perturbations of the metric of the fold (the size of the CMB anisotropies). Unit: dimensionless. Value: $\sim 10^{-5}$. Status: STANDARD.

The metric as a power series of terms of decreasing magnitude (the author's request): the order-0 term is the author's metric; the terms of order n in ϵ_r are the response of the bulk to the matter of the fold and of dirac16complex, which depends on σ; $\epsilon_\mathrm{c} \sim 10^{-5}$ (the size of the CMB anisotropies) counts the cosmological perturbations of the metric of the fold. The bookkeeping parameter ϵ_r is the ratio of the bulk to the fold gravity, $r = \kappa_4 V_*/(\kappa H)$ (FE-59). Each order obeys the linearisation of FE-1 and FE-4 around the previous orders, with the sources of that order. Status: DERIVED (the scheme).